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MIL-M-28787/335
19 November 1990

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

DIRECT MEMORY ACCESS MODULE, KEY CODE NBP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic direct memory access module.

1.2 Classification. This specification covers two classes of modules. Class I and class II are as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbooks. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 0.1000 failures per million operating hours for class I or 0.1000 failures per million operating hours for class II when predicted in accordance with MIL-HDBK-217. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be NBP.

3.6.2 Size. The module size shall be 1A Format A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned 1 through 40.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 7/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/ Min	Class I and II 25 ± 5°C 4/ Max	Class I 0 ± 5°C 4/ 60 ± 5°C 5/ Min	Class I 0 ± 5°C 4/ 60 ± 5°C 5/ Max	Class II -55 ± 5°C 4/ 85 ± 5°C 6/ Min	Class II -55 ± 5°C 4/ 85 ± 5°C 6/ Max	Units
A	Isolation, initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
B	Vcc stress	III	Q	---------Module shall pass all remaining tests---------						
C1-C24	Iih	IV	X	-10	10	-10	10	-10	10	uA
D1-D24	Iil	V	X	-10	10	-10	10	-10	10	uA
E1-E18	Voh	VI	X	2.4	4.5	2 4	4.5	2.4	4.5	V
E19	Voh	VI	X	3.3	4.5	3 3	4.5	3.3	4.5	V
E20-E27	Voh	VI	X	2.4	4.5	2.4	4.5	2.4	4.5	V
E28	Voh	VI	X	4.0	5.0	4 0	5.0	4.0	5.0	V
F1-F28	Vol	VII	X	0.0	0.45	0 0	0.45	0.0	0.45	V
G	Ioff	VIII	X	0	100	0	100	0	100	uA
H1-H20	Iozh	IX	X	-10	10	-10	10	-10	10	uA
J1-J20	Iozl	X	X	-10	10	-10	10	-10	10	uA
K	Icc	XI	X	0	130	0	150	0	175	mA
L1,L2	Functional	XII	X	----------See table XII----------						
M1,M2	Dynamic	XIII	S	----------See table XIII----------						
N	PSFC	XV	S	3.3	6.12	3 3	6.12	3.3	6.12	uF
P1-P28	Output crosstalk (high level)	XVI	Q	0	500	0	500	0	500	mV

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 7/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C 5/		Class II -55 ± 5°C 4/ 85 ± 5°C 6/		Units
				Min	Max	Min	Max	Min	Max	
Q1-Q28	Output crosstalk (low level)	XVII	Q	0	500	0	500	0	500	mV
R	PSTA	XVIII	Q	0	500	0	500	0	500	mV
S1-S9	Ci/o	XIX	Q	2	20	2	20	2	20	pF
S10-S24	Ci/o	XIX	Q	2	25	2	25	2	25	pF
S25-S31	Ci/o	XIX	Q	2	20	2	20	2	20	pF

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Values for 0°C and 60°C may be different. The first value given corresponds to the 0°C limit for that test condition. The second value given on the next line is the value for the 60°C limit.

6/ Values for -55°C and 85°C may be different. The first value given corresponds to the -55°C limit for that test condition. The second value given on the next line is the value for the 85°C limit.

7/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/335-1 for class I and M28787/335-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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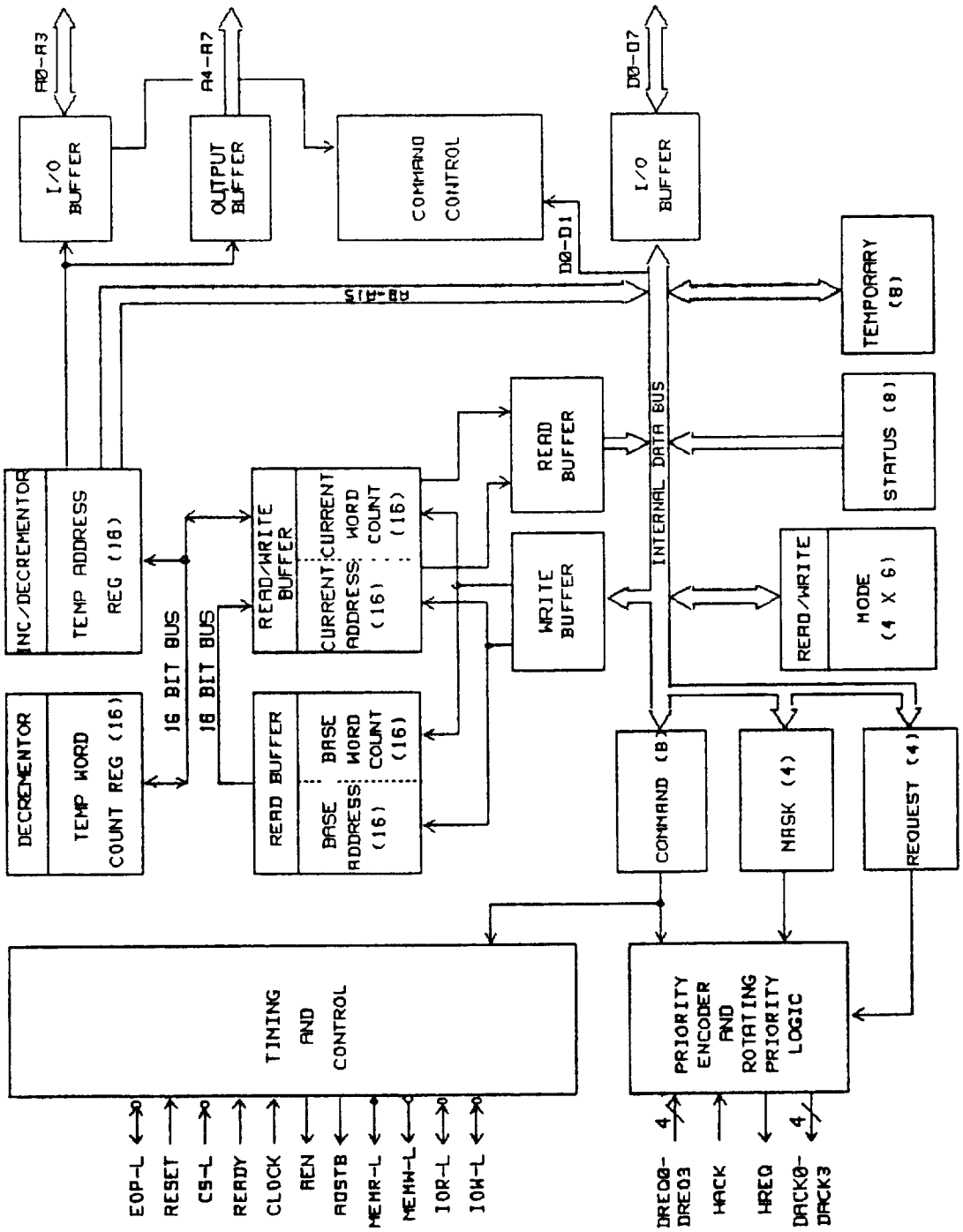


FIGURE 1. NBP functional block diagram.

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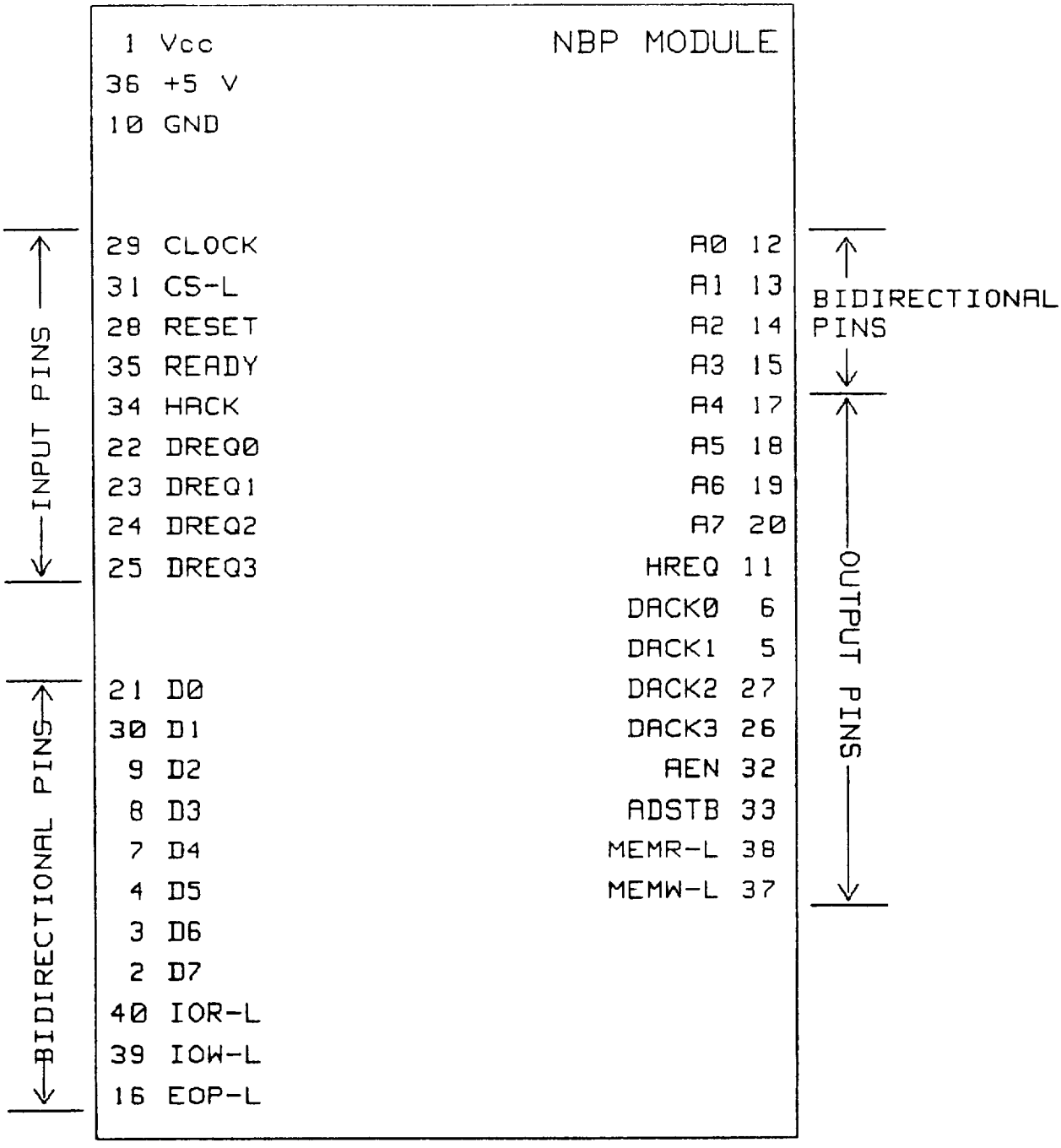


FIGURE 2. NBP pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through S shall be performed in any order with the following exceptions. Test conditions A and B are stress tests and shall be performed only at 25 ± 5°C. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A and B where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

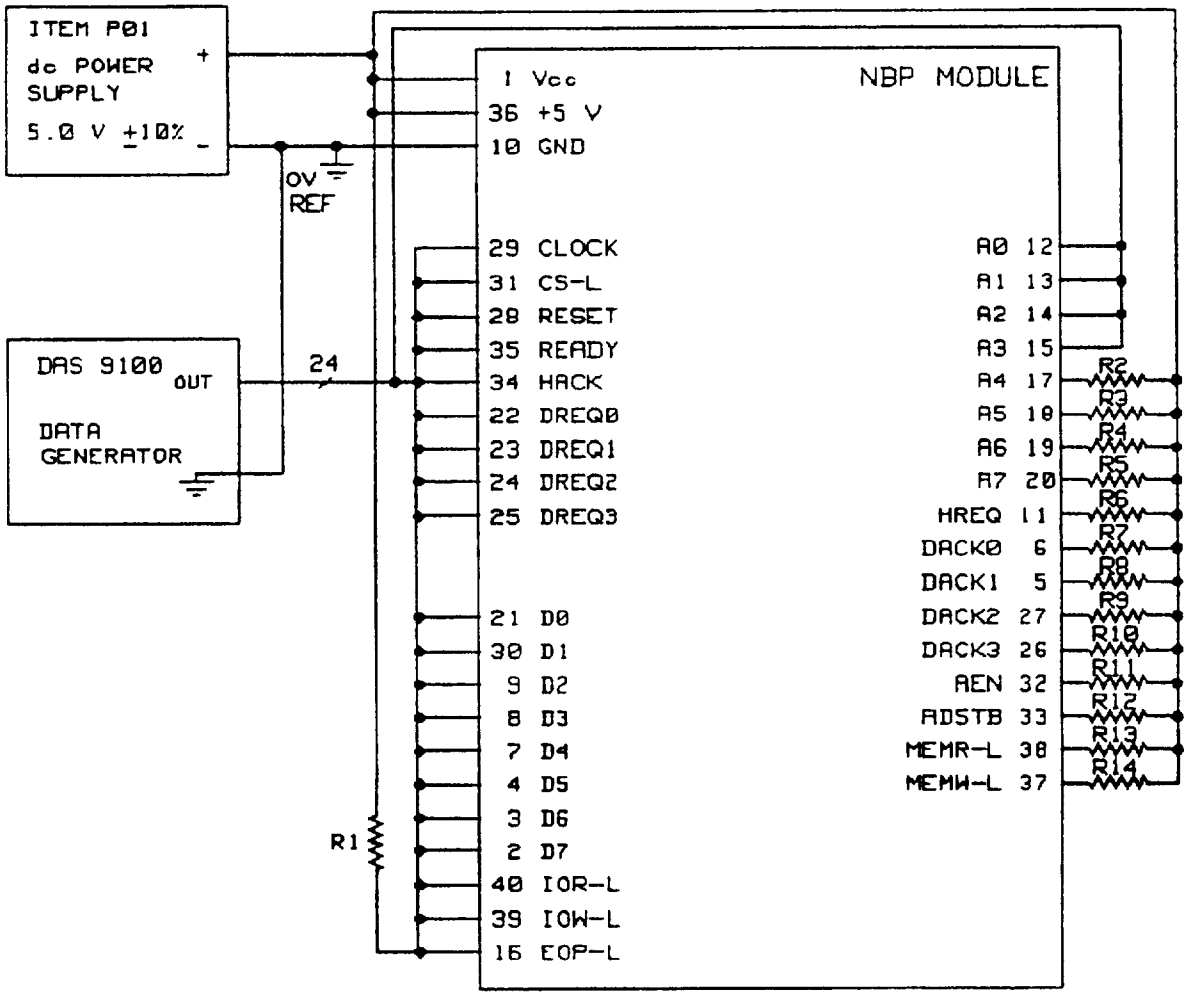
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through S in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through R shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition N, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTES:

1. R1 = 3.3 KOhm ± 5 percent, 1/4 W.

2. R2 - R14 = 1.62 KOhm ± 5 percent, 1/4 W.

3. Set the DAS 9100 to a pattern repetition rate of 1000 ± 200 nS, 50 percent duty cycle.

4. Pins 21,30,9,8,7,4,3,2,40,39,16,12,13,14 and 15 are bidirectional and each shall be loaded with a 1.62 KOhm ± 5 percent, 1/4 W resistor tied to Vcc when used as an output.

FIGURE 3. NBP operating temperature, component temperature and life test circuit.

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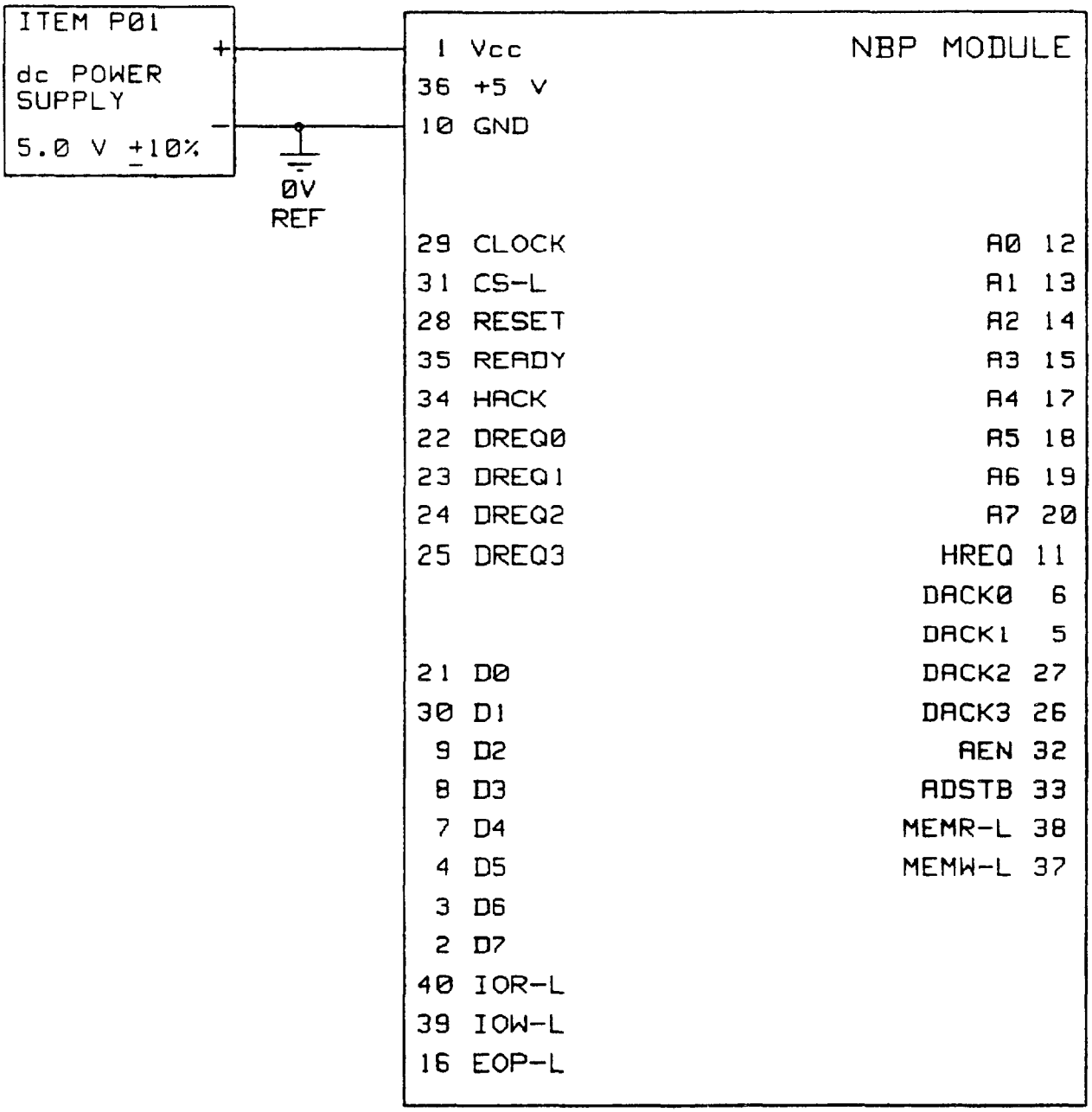


FIGURE 4. NBP vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests C through R shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests C through R shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ioff = output open drain leakage current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 0.963 W (max); 0.4813 W (typical).

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6.3.1 Input parameters.

Signal	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Cin max (pF)
CLOCK	0.8	2.0	-10.0	10	20
CS-L	0.8	2.0	-10.0	10	20
RESET	0.8	2.0	-10.0	10	20
READY	0.8	2.0	-10.0	10	20
HACK	0.8	2.0	-10.0	10	20
DREQ0	0.8	2.0	-10.0	10	20
DREQ1	0.8	2.0	-10.0	10	20
DREQ2	0.8	2.0	-10.0	10	20
DREQ3	0.8	2.0	-10.0	10	20
D0	0.8	2.0	-10.0	10	25
D1	0.8	2.0	-10.0	10	25
D2	0.8	2.0	-10.0	10	25
D3	0.8	2.0	-10.0	10	25

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6.3.1 Input parameters - Continued.

Signal	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Cin max (pF)
D4	0.8	2.0	-10.0	10	25
D5	0.8	2.0	-10.0	10	25
D6	0.8	2.0	-10.0	10	25
D7	0.8	2.0	-10.0	10	25
IOR-L	0.8	2.0	-10.0	10	25
IOW-L	0.8	2.0	-10.0	10	25
A0	0.8	2.0	-10.0	10	25
A1	0.8	2.0	-10.0	10	25
A2	0.8	2.0	-10.0	10	25
A3	0.8	2.0	-10.0	10	25
EOP-L	0.8	2.0	-10.0	10	25

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (uA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Cout max (pF)
A4	2.4	0.45	-200	3.2	-10	10	15.0
A5	2.4	0.45	-200	3.2	-10	10	15.0
A6	2.4	0.45	-200	3.2	-10	10	15.0
A7	2.4	0.45	-200	3.2	-10	10	15.0
MEMR-L	2.4	0.45	-200	3.2	-10	10	15.0
MEMW-L	2.4	0.45	-200	3.2	-10	10	15.0
HREQ	3.3	0.45	-100	3.2	NA	NA	15.0
DACK0	2.4	0.45	-200	3.2	NA	NA	NA
DACK1	2.4	0.45	-200	3.2	NA	NA	NA
DACK2	2.4	0.45	-200	3.2	NA	NA	NA
DACK3	2.4	0.45	-200	3.2	NA	NA	NA
AEN	2.4	0.45	-200	3.2	NA	NA	NA
ADSTB	2.4	0.45	-200	3.2	NA	NA	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (uA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Cout max (pF)
D0	2.4	0.45	-200	3.2	-10	10	25.0
D1	2.4	0.45	-200	3.2	-10	10	25.0
D2	2.4	0.45	-200	3.2	-10	10	25.0
D3	2.4	0.45	-200	3.2	-10	10	25.0
D4	2.4	0.45	-200	3.2	-10	10	25.0
D5	2.4	0.45	-200	3.2	-10	10	25.0
D6	2.4	0.45	-200	3.2	-10	10	25.0
D7	2.4	0.45	-200	3.2	-10	10	25.0
IOR-L	2.4	0.45	-200	3.2	-10	10	25.0
IOW-L	2.4	0.45	-200	3.2	-10	10	25.0
A0	2.4	0.45	-200	3.2	-10	10	25.0
A1	2.4	0.45	-200	3.2	-10	10	25.0
A2	2.4	0.45	-200	3.2	-10	10	25.0

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (uA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Cout max (pF)
A3	2.4	0.45	-200	3.2	-10	10	25.0
EOP-L	NA	0.45	NA	3.2	NA	NA	25.0

6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The module function is printed herein in the appendix.

6.4.1 Functional description of I/O pins.

Signal name	Pin number	I/O	Functional description
CLOCK	29	I	Clock. This input controls the internal operations of the NBP and its rate of data transfers. This input may be driven at up to 2 MHz.
CS-L	31	I	Chip Select. Chip Select is an active low input to select the NBP as an Input/Output (I/O) device during an I/O read or I/O write by the host Central Processing Unit (CPU). This allows CPU communication on the data bus. During multiple transfers to or from the NBP by the host CPU, CS-L may be held low provided that IOR-L or IOW-L is toggled following each transfer.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
RESET	28	I	Reset. Reset is an asynchronous active high input which clears the command, status, request and temporary registers. It also clears the first/last flip/flop and sets the mask register. Following a reset, the device is in the idle cycle.
READY	35	I	Ready. Ready is an input used to extend the memory read and write pulses from the NBP to accommodate slow memories or I/O peripheral devices.
HACK	34	I	Hold Acknowledge. The active high input from the CPU indicates that control of the system buses has been relinquished.
DREQ0 DREQ1 DREQ2 DREQ3	22 23 24 25	I I I I	Direct Memory Access (DMA) Request. The DMA request lines are asynchronous channel request inputs used by peripheral circuits to obtain DMA service. In fixed priority, DREQ0 has the highest priority and DREQ3 has lowest priority. Polarity of DREQ is a programmable. Reset initializes these lines to active high.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
D0 D1 D2 D3 D4 D5 D6 D7	21 30 9 8 7 4 3 2	I/O I/O I/O I/O I/O I/O I/O I/O	Data bus. The data bus lines are bi-directional three-state signals connected to the system data bus. The outputs are enabled during the I/O read by the host CPU, permitting the CPU to examine the contents of an address register, the status register, the temporary register or a word count register. The data bus is enabled to input data during a host CPU I/O write, allowing the CPU to program the NBP control registers. During DMA cycles the most significant eight bits of the address are output onto the data bus to be strobed into an external latch by ADSTB. In memory-to-memory operations, data from the source memory location comes into the NBP's temporary register on the read-to-memory half of the operation. On the write-to-memory half of the operation, the data bus outputs the temporary register data into the destination memory location.
IOR-L	40	I/O	I/O Read. I/O Read is a bidirectional active low three-state line. In the idle cycle, it is an input control signal used by the CPU to read the control registers. In the active cycle, it is an output control signal used by the NBP to access data from a peripheral during a DMA write transfer.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
IOW-L	39	I/O	I/O Write. I/O Write is a bidirectional active low three-state line. In the idle cycle it is an input control signal used by the CPU to load information into the NBP. In the active cycle it is an output control signal used by the NBP to load data to the peripheral during DMA. Write operations by the CPU to the NBP require a rising IOW-L edge following each data byte transfer. It is not sufficient to hold the IOW-L pin low and toggle CS-L.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
EOP-L	16	I/O	End Of Process. EOP-L is an active low bidirectional open-drain signal providing information concerning the completion of DMA service. When a channel's word count goes to zero, the NBP pulses EOP-L low to provide the peripheral with a completion signal. EOP-L may also be pulled by the peripheral to cause premature completion. The reception of EOP-L, either internal or external, causes the currently active channel to terminate the service, to set its Terminal Count (TC) bit in the status register and to reset its request bit. If autoinitialization is selected for the channel, the current registers will be updated from the base registers. Otherwise, the channel's mask bit will be set and the register contents will remain unaltered. During memory-to-memory transfers, EOP-L will be output when the TC for channel 1 occurs. EOP-L always applies to the channel with an active DACK; external EOP-Ls are disregarded when DACK0-DACK3 are all inactive. Because EOP-L is an open-drain signal, an external pull-up resistor is required. Values of 3.3 KOhm or 4.7 KOhm are recommended; the EOP-L pin cannot sink the current passed by a 1 KOhm pull-up.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
A0 A1 A2 A3	12 13 14 15	I/O I/O I/O I/O	Address bus, Least Significant Bits (LSB). The four least significant address lines are bidirectional three-state signals. During the DMA idle cycles they are inputs and allow the host CPU to load or read control registers. When the DMA is active, they are outputs and provide the lower four bits of the output address.
A4 A5 A6 A7	17 18 19 20	O O O O	Address bus, Most Significant Bits (MSB). The four most significant address lines are three-state outputs and provide four bits of address. These lines are enabled only during DMA service.
HREQ	11	O	Hold Request. The Hold Request to the CPU is used by the DMA to request control of the system bus. Software requests or unmasked DREQs cause the NBP to issue HREQ. HREQ is an active high signal.
DACK0 DACK1 DACK2 DACK3	6 5 27 26	O O O O	DMA Acknowledge. The DMA Acknowledge lines indicate that a channel is active. In many systems they will be used to select a peripheral. Only one DACK will be active at a time and none will be active unless the DMA is in control of the bus. The polarity of the lines is programmable. Reset initializes them to active low.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
AEN	32	O	Address Enable. Address Enable is an active high signal used to disable the system bus during DMA cycles to enable the output of the external latch which holds the upper byte of the address. Note that during DMA transfers HACK and AEN should be used to deselect all other I/O peripherals during the DMA operation. The NBP automatically deselects itself by disabling the CS-L input during DMA transfers.
ADSTB	33	O	Address Strobe. The active high Address Strobe is used to strobe the upper address byte from D0-D7 into an external latch.
MEMR-L	38	O	Memory Read. The Memory Read signal is an active low three-state output used to access data from the selected memory location during a memory-to-peripheral or a memory-to-memory transfer.
MEMW-L	37	O	Memory Write. The Memory Write signal is an active low three-state output used to write data to the selected memory location during a peripheral-to-memory or a memory-to-memory transfer.
+5V	36	I	This pin is an input that should always be at a logic high level. An internal pull-up resister will establish a logic high when the pin is left floating. Alternatively, this may be tied to Vcc.

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6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Vcc stress test - table III.

(c) Iih test - table IV.

(d) Iil test - table V.

(e) Voh test - table VI, figure 5.

(f) Vol test - table VII, figure 5.

(g) Ioff test - table VIII.

(h) Iozh test - table IX.

(i) Iozl test - table X.

(j) Icc test - table XI.

(k) Functional test - table XII, figure 5.

(l) Dynamic test - tables XIII and XIV, figures 5, 6 and 7.

(m) PSFC test - table XV.

(n) Output crosstalk tests - tables XVI and XVII, figures 5, 8 and 9.

(o) PSTA test - table XVIII, figures 10 and 11.

(p) Ci/o test - table XIX.

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6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0066)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation	All pins	Frame	10	INF	MOhm	1-7

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A A

2. The megohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
B	Vcc	7.0	1,31	10,15 25,40	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	265 mA	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table III.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

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TABLE IV. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
C1	CLOCK	29	5.5	-10	10	uA	1-3
C2	CS-L	31					
C3	RESET	28					
C4	READY	35					
C5	HACK	34					
C6	DREQ0	22					
C7	DREQ1	23					
C8	DREQ2	24					
C9	DREQ3	25					
C10	D0	21					
C11	D1	30					
C12	D2	9					
C13	D3	8					
C14	D4	7					
C15	D5	4					
C16	D6	3					
C17	D7	2					
C18	IOR-L	40					
C19	IOW-L	39					
C20	EOP-L	16	5.5	-10	10	uA	1-3

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TABLE IV. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
C21	A0	12	5.5	-10	10	uA	1-3
C22	A1	13					
C23	A2	14					
C24	A3	15	5.5	-10	10	uA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	C1-C24
Power supply current limit	220 mA	NA	C1-C24
Forcing voltage, (Vih)	5.5 V	± 55 mV	C1-C24
Pattern voltage high, (VH)	3.0 V	± 100 mV	C1-C24
low, (VL)	0.0 V	± 100 mV	C1-C24
Input current measurement	NA	± 1 uA	C1-C24

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IV and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

(e) All floating inputs should be tied to ground, except pin 36.

3. Abbreviations used are defined in 6.1.

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TABLE V. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D1	CLOCK	29	0.4	-10	10	uA	1-3
D2	CS-L	31					
D3	RESET	28					
D4	READY	35					
D5	HACK	34					
D6	DREQ0	22					
D7	DREQ1	23					
D8	DREQ2	24					
D9	DREQ3	25					
D10	D0	21					
D11	D1	30					
D12	D2	9					
D13	D3	8					
D14	D4	7					
D15	D5	4					
D16	D6	3					
D17	D7	2					
D18	IOR-L	40					
D19	IOW-L	39					
D20	EOP-L	16	0.4	-10	10	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:51Z
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TABLE V. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	A0	12	0.4	-10	10	uA	1-3
D22	A1	13					
D23	A2	14					
D24	A3	15	0.4	-10	10	uA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D24
Power supply current limit	220 mA	NA	D1-D24
Forcing voltage, (Vil)	0.4 V	± 20 mV	D1-D24
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	D1-D24 D1-D24
Input current measurement	NA	± 1 uA	D1-D24

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

(e) All floating inputs should be tied to 3V, except pin 36.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E1	D0	21	-0.2	2.4	4.5	V	1-4
E2	D1	30					
E3	D2	9					
E4	D3	8					
E5	D4	7					
E6	D5	4					
E7	D6	3					
E8	D7	2					
E9	IOR-L	40					
E10	IOW-L	39					
E11	A0	12					
E12	A1	13					
E13	A2	14					
E14	A3	15					
E15	A4	17					
E16	A5	18					
E17	A6	19					
E18	A7	20	-0.2	2.4	4.5	V	1-4
E19	HREQ	11	-0.1	3.3	4.5	V	1-4
E20	DACK0	6	-0.2	2.4	4.5	V	1-4

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TABLE VI. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	DACK1	5	-0.2	2.4	4.5	V	1-4
E22	DACK2	27					
E23	DACK3	26					
E24	AEN	32					
E25	ADSTB	33					
E26	MEMR-L	38					
E27	MEMW-L	37	-0.2	2.4	4.5	V	1-4
E28	+5V	36	NA	4.0	5.0	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	E1-E28
Power supply current limit	220 mA	NA	E1-E28
Forcing current, (Ioh)	-0.2 -0.1 -0.2 NA	± 2 uA ± 1 uA ± 2 uA NA	E1-E18 E19 E20-E27 E28
Pattern voltage high, (VH) low, (VL)	2.0 V 0.8 V	± 20 mV ± 8 mV	E1-E28 E1-E28

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage measurement	NA NA NA NA	± 105 mV ± 60 mV ± 105 mV ± 50 mV	E1-E18 E19 E20-E27 E28

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

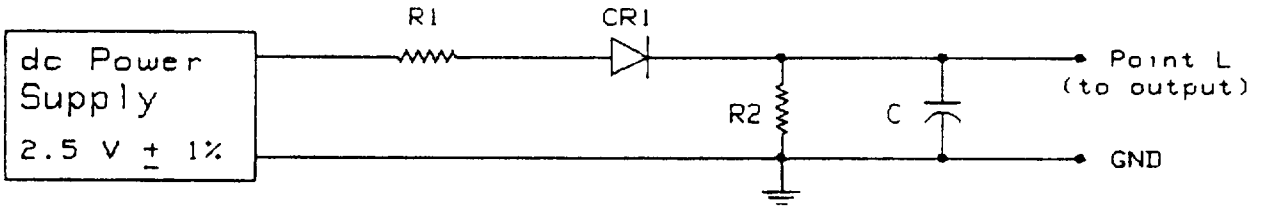
(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

(e) For test condition E28, no forcing current is applied. The voltage is measured with the pin floating.

3. The load shall be as indicated on figure 5.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

(c) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 5. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are listed below.

4. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,3,4,5,6,7,8,9, 12,13,14,15,17,18, 19,20,21,26,27,30, 32,33,37,38,39,40	12100	3.2	0.45
11	33200	3.2	0.45
16	3320	3.2	0.45

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE VII. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F1	D0	21	3.2	0.0	0.45	V	1-4
F2	D1	30					
F3	D2	9					
F4	D3	8					
F5	D4	7					
F6	D5	4					
F7	D6	3					
F8	D7	2					
F9	IOR-L	40					
F10	IOW-L	39					
F11	EOP-L	16					
F12	A0	12					
F13	A1	13					
F14	A2	14					
F15	A3	15					
F16	A4	17					
F17	A5	18					
F18	A6	19					
F19	A7	20					
F20	HREQ	11	3.2	0.0	0.45	V	1-4

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TABLE VII. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	DACK0	6	3.2	0.0	0.45	V	1-4
F22	DACK1	5					
F23	DACK2	27					
F24	DACK3	26					
F25	AEN	32					
F26	ADSTB	33					
F27	MEMR-L	38					
F28	MEMW-L	37	3.2	0.0	0.45	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	F1-F28
Power supply current limit	220 mA	NA	F1-F28
Forcing current, (Iol)	3.2 mA	± 160 uA	F1-F28
Pattern voltage high, (VH) low, (VL)	2.0 V 0.8 V	± 20 mV ± 8 mV	F1-F28 F1-F28

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage measurement	NA	± 22.5 mV	F1-F28

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VII and the voltage then measured.

3. The load shall be as indicated on figure 5.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Output leakage current, open drain, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G	EOP-L	16	5.5	0	100	uA	1-3

NOTES:

1. Output leakage current, open drain, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G
Power supply current limit	220 mA	NA	G
Forcing voltage, (Vol)	5.5 V	± 55 mV	G
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	G G
Output current measurement	NA	± 5 uA	G

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	D0	21	2.4	-10	10	uA	1-3
H2	D1	30					
H3	D2	9					
H4	D3	8					
H5	D4	7					
H6	D5	4					
H7	D6	3					
H8	D7	2					
H9	IOR-L	40					
H10	IOW-L	39					
H11	A0	12					
H12	A1	13					
H13	A2	14					
H14	A3	15					
H15	A4	17					
H16	A5	18					
H17	A6	19					
H18	A7	20					
H19	MEMR-L	38					
H20	MEMW-L	37	2.4	-10	10	uA	1-3

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NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H20
Power supply current limit	220 mA	NA	H1-H20
Forcing voltage, (Voh)	2.4 V	± 120 mV	H1-H20
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H20 H1-H20
Output current measurement	NA	± 1 uA	H1-H20

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	D0	21	0.4	-10	10	uA	1-3
J2	D1	30					
J3	D2	9					
J4	D3	8					
J5	D4	7					
J6	D5	4					
J7	D6	3					
J8	D7	2					
J9	IOR-L	40					
J10	IOW-L	39					
J11	A0	12					
J12	A1	13					
J13	A2	14					
J14	A3	15					
J15	A4	17					
J16	A5	18					
J17	A6	19					
J18	A7	20					
J19	MEMR-L	38					
J20	MEMW-L	37	0.4	-10	10	uA	1-3

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NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	J1-J20
Power supply current limit	220 mA	NA	J1-J20
Forcing voltage, (Vol)	0.4 V	± 20 mV	J1-J20
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J20 J1-J20
Output current measurement	NA	± 1 uA	J1-J20

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table X and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XI. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
K	Icc	1	0	130	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K
Power supply current limit	220 mA	NA	K
Pattern voltage high, (VH)	3.0 V	± 100 mV	K
low, (VL)	0.0 V	± 100 mV	K
Power supply current measurement	NA	± 6.5 mA	K

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XI.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Functional test.

Test cond	Test description	Notes
L1	Functional test minimum power supply voltage maximum pattern rate	1-3
L2	Functional test maximum power supply voltage maximum pattern rate	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	L1 L2
Power supply current limit	220 mA	NA	L1,L2
Pattern voltage high, (VH) low, (VL)	 2.0 V 0.8 V	 ± 20 mV ± 8 mV	 L1,L2 L1,L2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105.0 mV ± 155.0 mV ± 22.5 mV	 L1 L2 L1,L2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XII using only a capacitive load of 100 pF (see figure 5). The output compare levels shall be as follows:

High level: 2.4 V $\leq$ Voh $\leq$ Vcc
High level: 3.3 V $\leq$ Voh $\leq$ Vcc (HREQ)
Low level: 0.0 V $\leq$ Vol $\leq$ 0.45 V

(c) The high level verification accuracies for pin 11 (HREQ) are $\pm$ 60 mV for test condition L1 and $\pm$ 110 mV for test condition L2.

(d) Timing information appears with the dynamic test.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Dynamic test.

Test cond	Test description	Notes
M1	Dynamic test, sample maximum pattern rate minimum power supply	1-8
M2	Dynamic test, sample maximum pattern rate maximum power supply	1-8

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	M1 M2
Power supply current limit	220 mA	NA	M1,M2
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 M1,M2 M1,M2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105.0 mV ± 155.0 mV ± 22.5 mV	 M1 M2 M1,M2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: 2.4 V $\leq$ Voh $\leq$ Vcc
High level: 3.3 V $\leq$ Voh $\leq$ Vcc (HREQ)
Low level: 0.0 V $\leq$ Vol $\leq$ 0.45 V

(c) The high level verification accuracies for pin 11 (HREQ) are $\pm$ 60 mV for test condition M1 and $\pm$ 110 mV for test condition M2.

3. The load circuit for this test is specified on figure 5.

4. All of the input waveforms shall meet the requirements shown on figure 6.

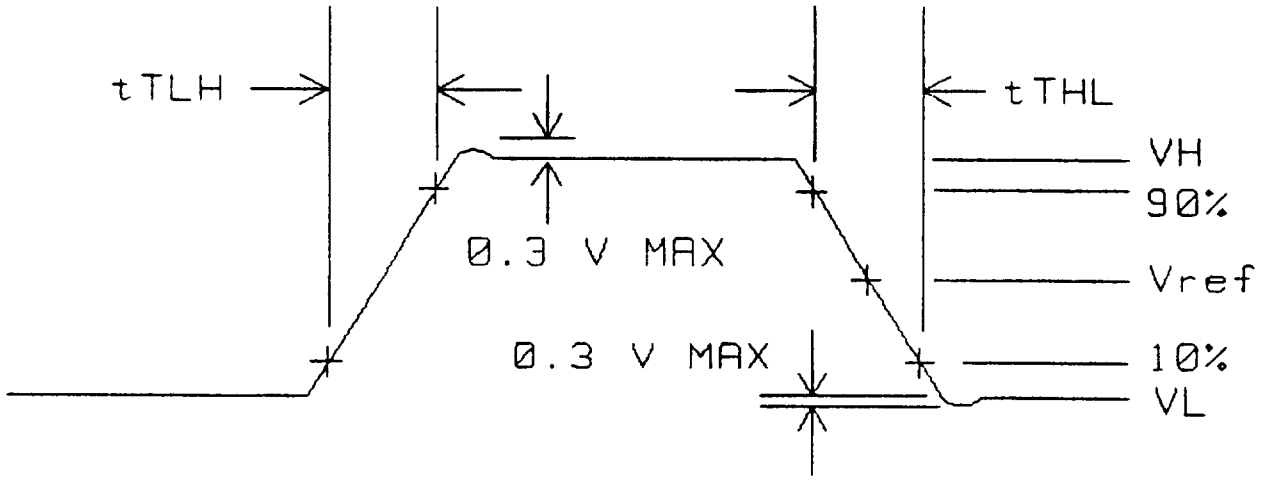
5. This test shall be performed as indicated in table XIII. The dynamic parameters are found in table XIV.

6. I/O waveforms are as referenced on figure 7.

7. Test conditions M1 and M2 shall be run twice. The load for pin 11 will change as follows for the second run: R2 = 3.32 KOhm $\pm$ 1 percent, 1/8 W and R3 (resistor between Vcc and pin 11) = 7.15 KOhm $\pm$ 1 percent, 1/8 W. Then select R1 as indicated on figure 5.

8. Abbreviations used are defined in 6.1.

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NOTES:

1. VH and VL are as specified in the dynamic test tolerance table.

2. Vref = 1.5 V.

3. Rise and fall times for the above waveform shall be 20 nS maximum.

FIGURE 6. Sample input waveform for table XIII.

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TABLE XIV. Dynamic parameters.

Ref num	Symbol	Parameter	Class I						Class II						Units	Fig
			Minimum			Maximum			Minimum			Maximum				
			0°C	25°C	60°C	0°C	25°C	60°C	-55°C	25°C	85°C	-55°C	25°C	85°C		
1	PRP	CLOCK cycle time	320	320	320	-	-	-	320	320	320	-	-	-	nS	7c
2	PWH	CLOCK high time	120	120	120	-	-	-	120	120	120	-	-	-	nS	7c
3	PWL	CLOCK low time	150	150	150	-	-	-	150	150	150	-	-	-	nS	7c
4	tTLH	CLOCK transition time high	-	-	-	10	10	10	-	-	-	10	10	10	nS	7c,7d 7e,7f
5	tTHL	CLOCK transition time low	-	-	-	10	10	10	-	-	-	10	10	10	nS	7c,7d 7e,7f
6	tPD	AEN high from CLOCK low (S1)	-	-	-	310	310	310	-	-	-	310	310	310	nS	7c
7	tPD	AEN low from CLOCK high (S1)	-	-	-	210	210	210	-	-	-	210	210	210	nS	7c
11	tH	A0-A7 from READ-L high hold time	PRP-100	PRP-100	PRP-100	-	-	-	PRP-100	PRP-100	PRP-100	-	-	-	nS	7c
12	tH	D0-D7 from ADSTB low hold time	50	50	50	-	-	-	50	50	50	-	-	-	nS	7c
13	tH	A0-A7 from WRITE-L high hold time	PRP-50	PRP-50	PRP-50	-	-	-	PRP-50	PRP-50	PRP-50	-	-	-	nS	7c
14	tPD	DACK0-DACK3 valid from CLOCK low delay time	-	-	-	290	290	290	-	-	-	290	290	290	nS	7c
15	tPD	INT EOP-L high from CLOCK high delay time	-	-	-	260	260	260	-	-	-	260	260	260	nS	7c,7d
16	tPD	INT EOP-L low to CLOCK high delay time	-	-	-	260	260	260	-	-	-	260	260	260	nS	7c,7d

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TABLE XIV. Dynamic parameters - Continued

Ref num	Symbol	Parameter	Class I						Class II						Units	Fig
			Minimum			Maximum			Minimum			Maximum				
			0°C	25°C	60°C	0°C	25°C	60°C	-55°C	25°C	85°C	-55°C	25°C	85°C		
17	tPD	A0-A7 stable from CLOCK high	-	-	-	260	280	260	-	-	-	260	260	260	nS	7c,7d
18	tS	D0-D7 to ADSTB low setup time	100	100	100	-	-	-	100	100	100	-	-	-	nS	7c
19	tPD	CLOCK high to READ-L or WRITE-L low delay 1/	-	-	-	280	280	280	-	-	-	280	280	280	nS	7c,7d 7f
20	tPD	READ-L high from CLOCK high (S4) delay time 1/	-	-	-	290	290	290	-	-	-	290	290	290	nS	7c,7d 7f
21	tPD	WRITE-L high from CLOCK high (S4) delay time 1/	-	-	-	210	210	210	-	-	-	210	210	210	nS	7c,7d 7f
22a	tPD	HREQ valid from CLOCK high delay time 7/	-	-	-	170	170	170	-	-	-	170	170	170	nS	7c
22b	tPD	HREQ valid from CLOCK high delay time 7/	-	-	-	260	260	260	-	-	-	260	260	260	nS	7c
23	tS	EXT EOP-L low from CLOCK low setup time	60	60	60	-	-	-	60	60	60	-	-	-	nS	7c,7d
24	PWLEOP	EXT EOP-L pulse width	300	300	300	-	-	-	300	300	300	-	-	-	nS	7c,7d
25	tPZV	A0-A7 float to active delay from CLOCK high	-	-	-	260	260	260	-	-	-	260	260	260	nS	7c,7d

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TABLE XIV Dynamic parameters - Continued

Ref num	Symbol	Parameter	Class I						Class II						Units	Fig
			Minimum			Maximum			Minimum			Maximum				
			0°C	25°C	60°C	0°C	25°C	60°C	-55°C	25°C	85°C	-55°C	25°C	85°C		
26	tPD	READ-L or WRITE-L active from CLOCK high	-	-	-	210	210	210	-	-	-	210	210	210	nS	7c,7d
27	tPZV	D0-D7 float to active delay from CLOCK high	-	-	-	310	310	310	-	-	-	310	310	310	nS	7c,7d
28	tS	HACK valid to CLOCK high setup time	100	100	100	-	-	-	100	100	100	-	-	-	nS	7c
29	tH	Input data from MEMR-L high hold time	0	0	0	-	-	-	0	0	0	-	-	-	nS	7d
30	tS	Input data to MEMR-L high setup time	250	250	250	-	-	-	250	250	250	-	-	-	nS	7d
31	tH	Output data from MEMW-L high hold time	20	20	20	-	-	-	20	20	20	-	-	-	nS	7d
32	tPD	Output data valid to MEMW-L high 6/	200	200	200	-	-	-	200	200	200	-	-	-	nS	7d
33	tS	DREQ0-DREQ3 to CLOCK low (S1,S4) setup time	120	120	120	-	-	-	120	120	120	-	-	-	nS	7c
34	tH	CLOCK to READY low hold time	20	20	20	-	-	-	20	20	20	-	-	-	nS	7e,7f
35	tS	READY to CLOCK low setup time	100	100	100	-	-	-	100	100	100	-	-	-	nS	7e,7f

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TABLE XIV. Dynamic parameters - Continued.

Ref num	Symbol	Parameter	Class I						Class II						Units	Fig
			Minimum			Maximum			Minimum			Maximum				
			0°C	25°C	60°C	0°C	25°C	60°C	-55°C	25°C	85°C	-55°C	25°C	85°C		
36	tPD	ADST0 high from CLOCK high delay time	-	-	-	210	210	210	-	-	-	210	210	210	nS	7c,7d
37	tPD	ADST0 low from CLOCK high delay time	-	-	-	150	150	150	-	-	-	150	150	150	nS	7c,7d
38	tS	A0-A3 valid or CS-L low to READ-L low	50	50	50	-	-	-	50	50	50	-	-	-	nS	7b
39	tS	A0-A3 valid to WRITE-L high setup time	200	200	200	-	-	-	200	200	200	-	-	-	nS	7a
40	tS	CS-L low to WRITE-L high setup time	200	200	200	-	-	-	200	200	200	-	-	-	nS	7a
41	tS	Data valid to WRITE-L high setup time	200	200	200	-	-	-	200	200	200	-	-	-	nS	7a
42	tH	A0-A3 or CS-L hold from READ-L high	0	0	0	-	-	-	0	0	0	-	-	-	nS	7b
43	tPD	Data access from READ-L low	-	-	-	310	310	310	-	-	-	310	310	310	nS	7b
46	tS	Reset to first IOW-L or IOR-L	2PRP	2PRP	2PRP	-	-	-	2PRP	2PRP	2PRP	-	-	-	-	7g
47	PWHRES	Reset pulse width	300	300	300	-	-	-	300	300	300	-	-	-	nS	7g
48	PWLRD	READ-L width	300	300	300	-	-	-	300	300	300	-	-	-	nS	7b

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TABLE XIV. Dynamic parameters - Continued

Ref num	Symbol	Parameter	Class I						Class II						Units	Fig
			Minimum			Maximum			Minimum			Maximum				
			0°C	25°C	60°C	0°C	25°C	60°C	-55°C	25°C	85°C	-55°C	25°C	85°C		
49	tH	A0-A3 from WRITE-L high hold time	20	20	20	-	-	-	20	20	20	-	-	-	nS	7a
50	tH	CS-L high from WRITE-L high hold time	20	20	20	-	-	-	20	20	20	-	-	-	nS	7a
51	tH	Data from WRITE-L high hold time	30	30	30	-	-	-	30	30	30	-	-	-	nS	7a
52	PWLWR	Write width	200	200	200	-	-	-	200	200	200	-	-	-	nS	7a
53	tH	DREQ from DACK valid hold time	0	0	0	-	-	-	0	0	0	-	-	-	nS	7c
54	tPD	HREQ to HACK delay time	1PRP	1PRP	1PRP	-	-	-	1PRP	1PRP	1PPP	-	-	-	nS	7c

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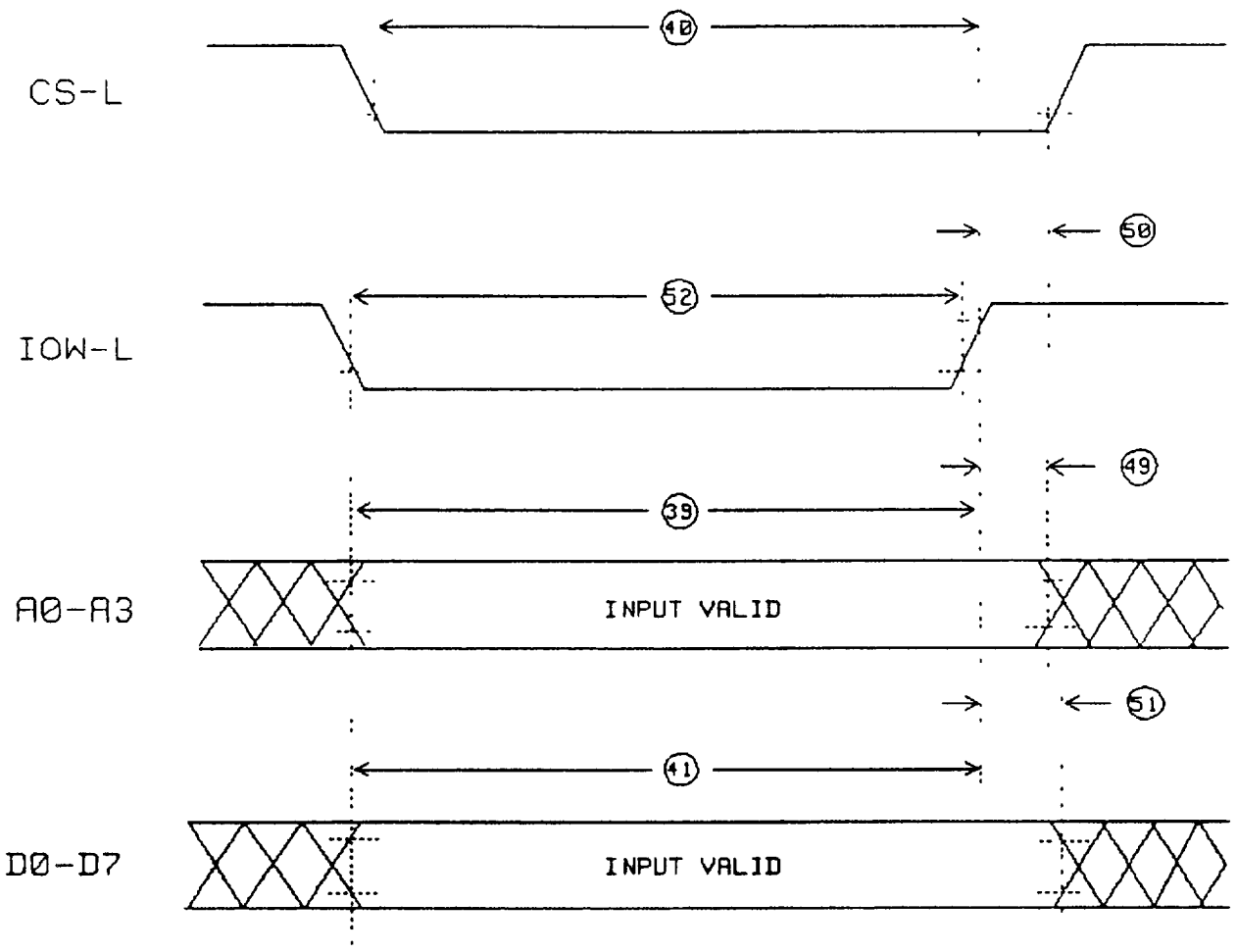


FIGURE 7a. Dynamic waveforms for table XIV.

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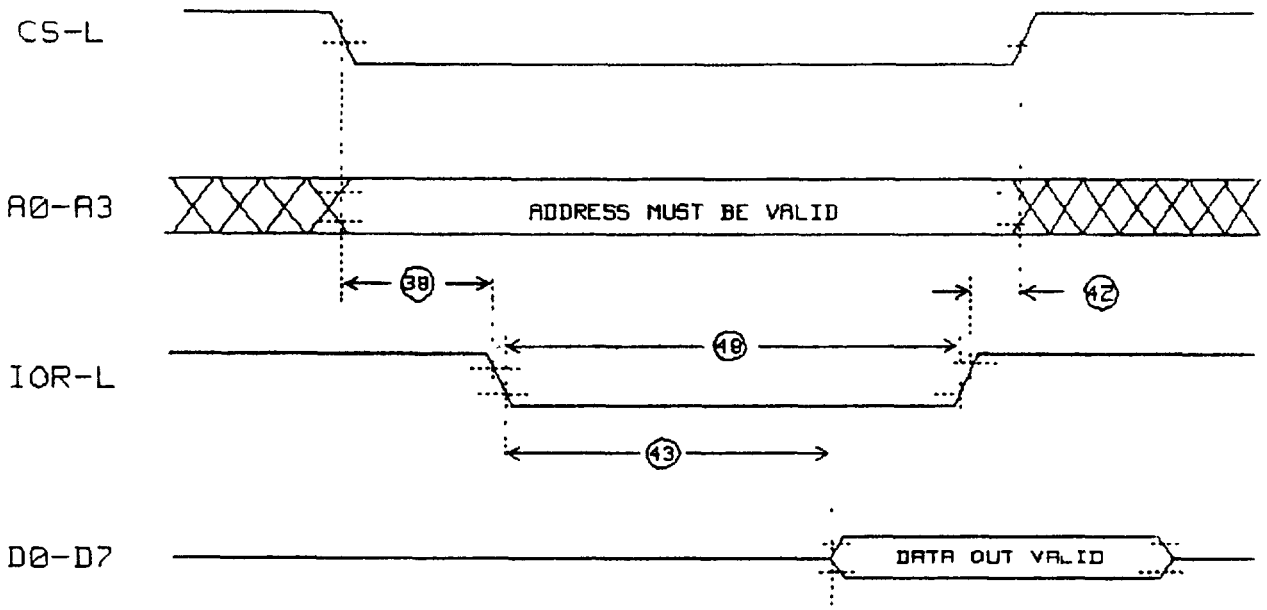


FIGURE 7b. Dynamic waveforms for table XIV.

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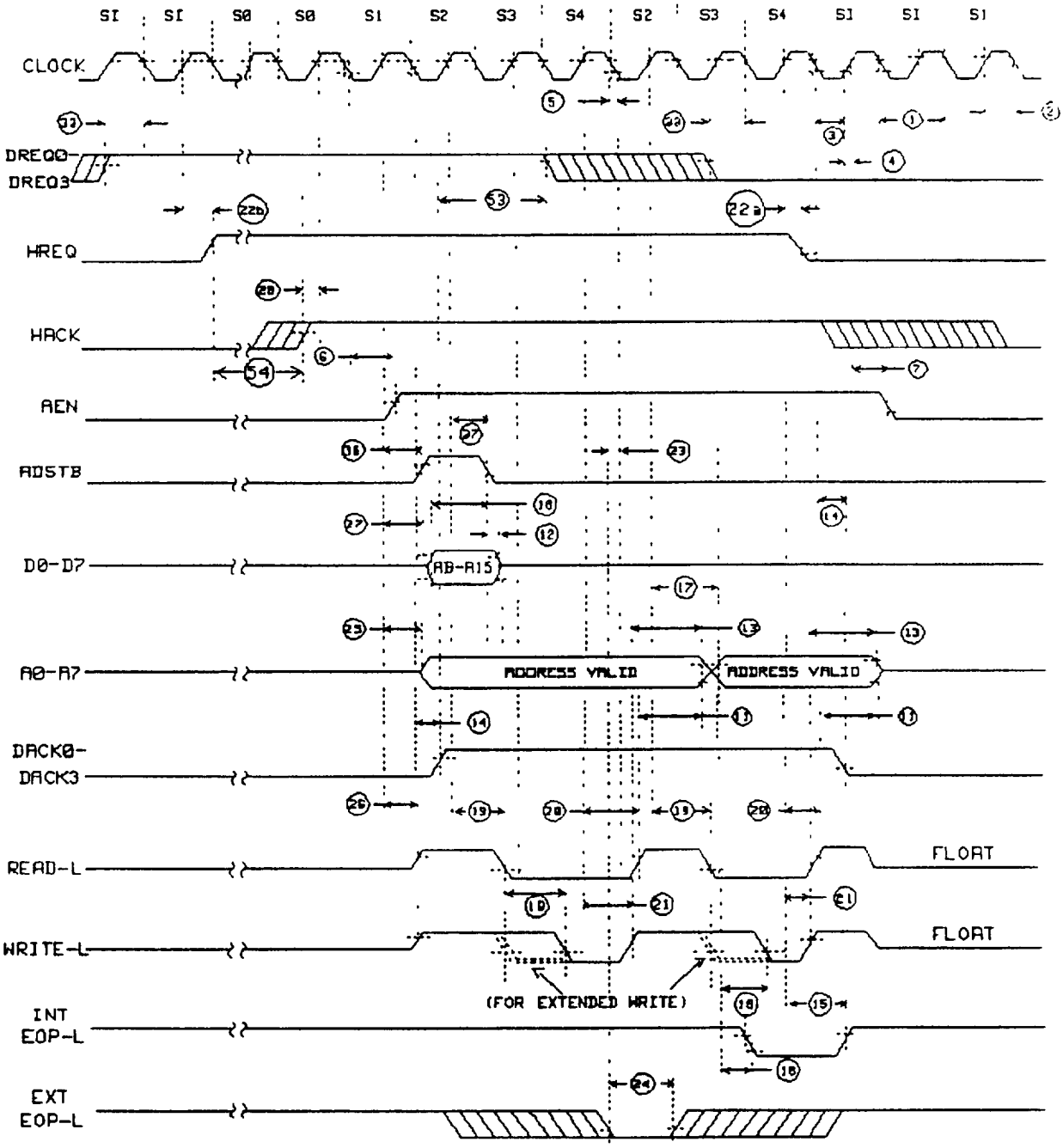


FIGURE 7c. Dynamic waveforms for table XIV.

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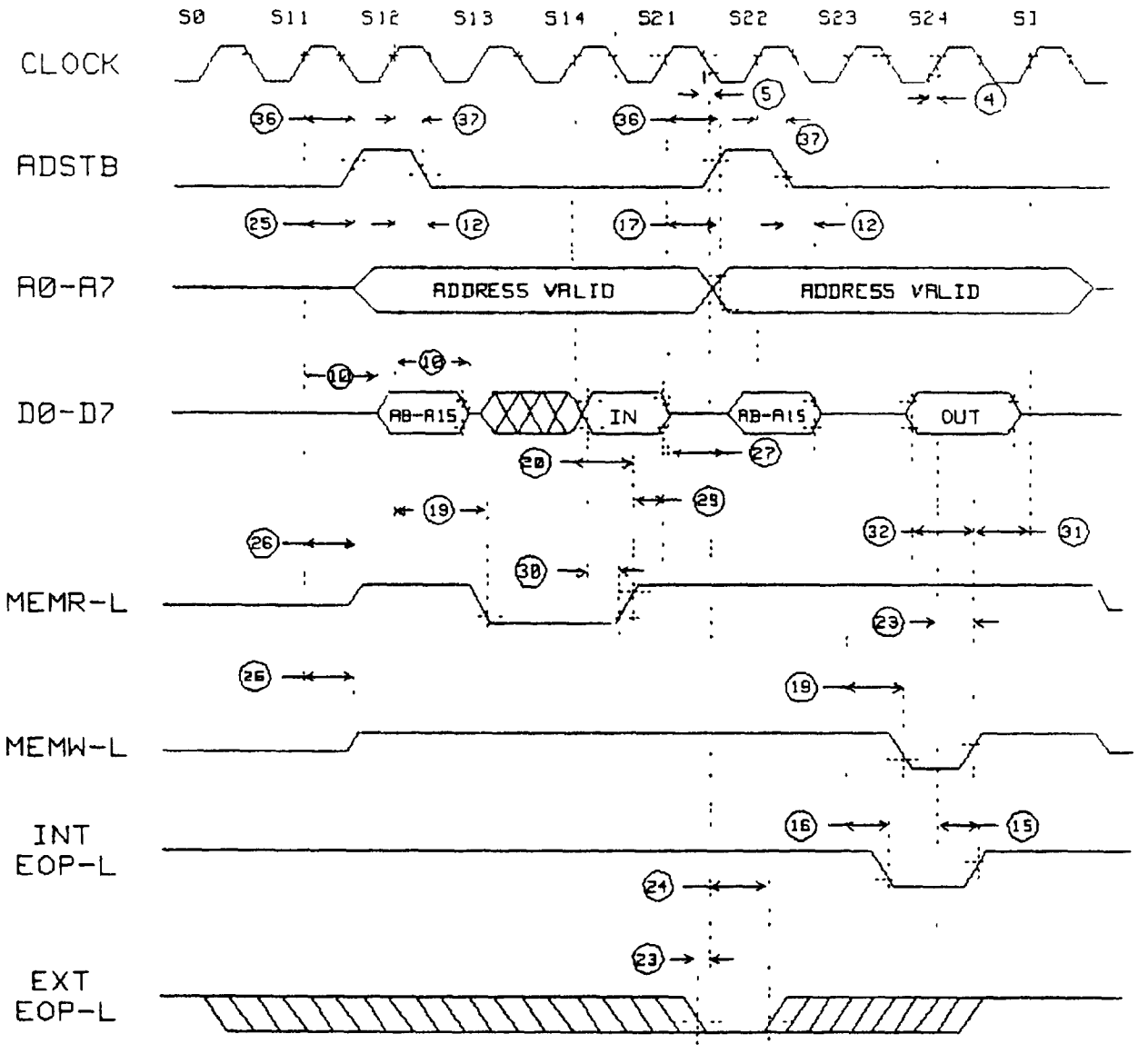


FIGURE 7d. Dynamic waveforms for table XIV.

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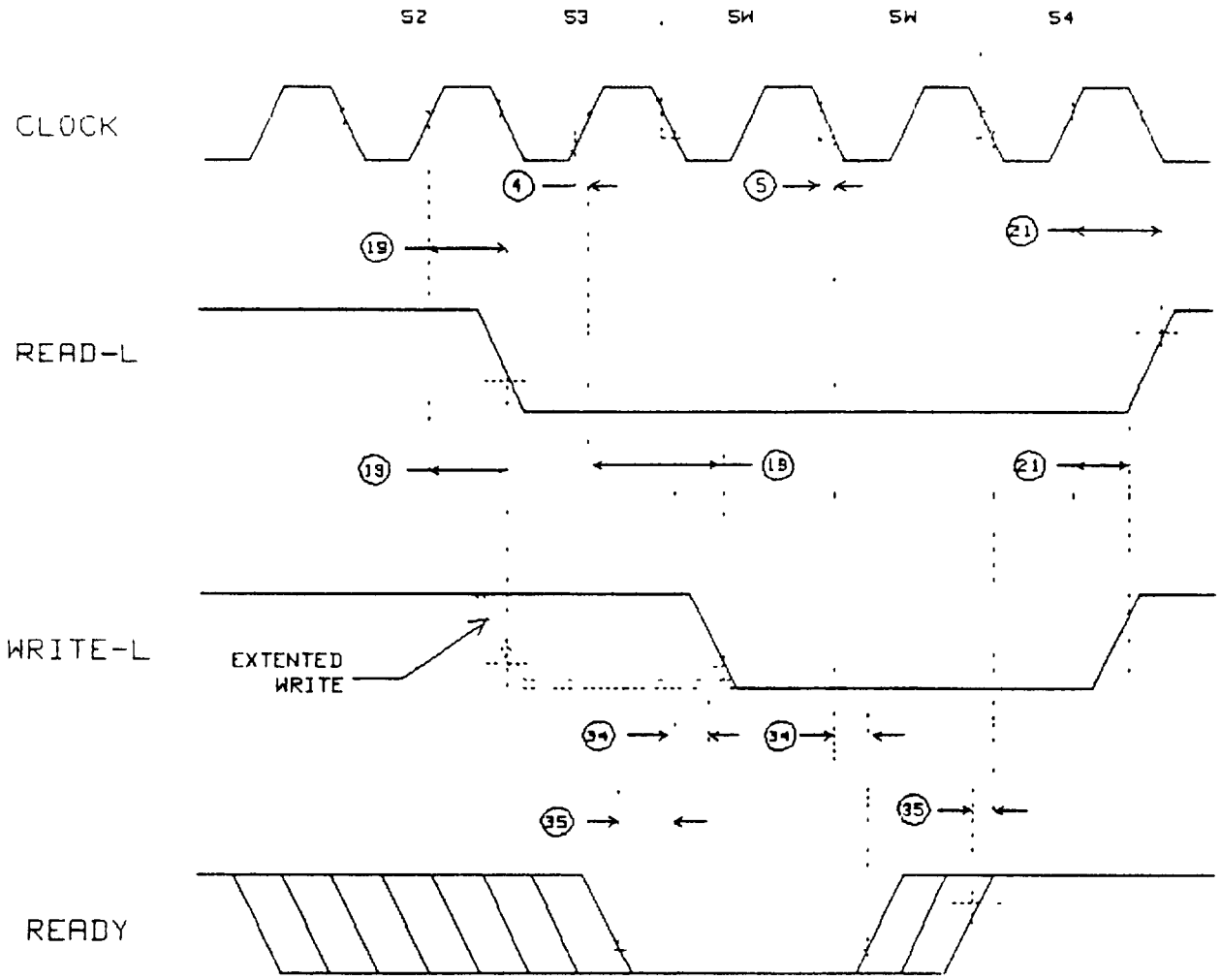


FIGURE 7e. Dynamic waveforms for table XIV.

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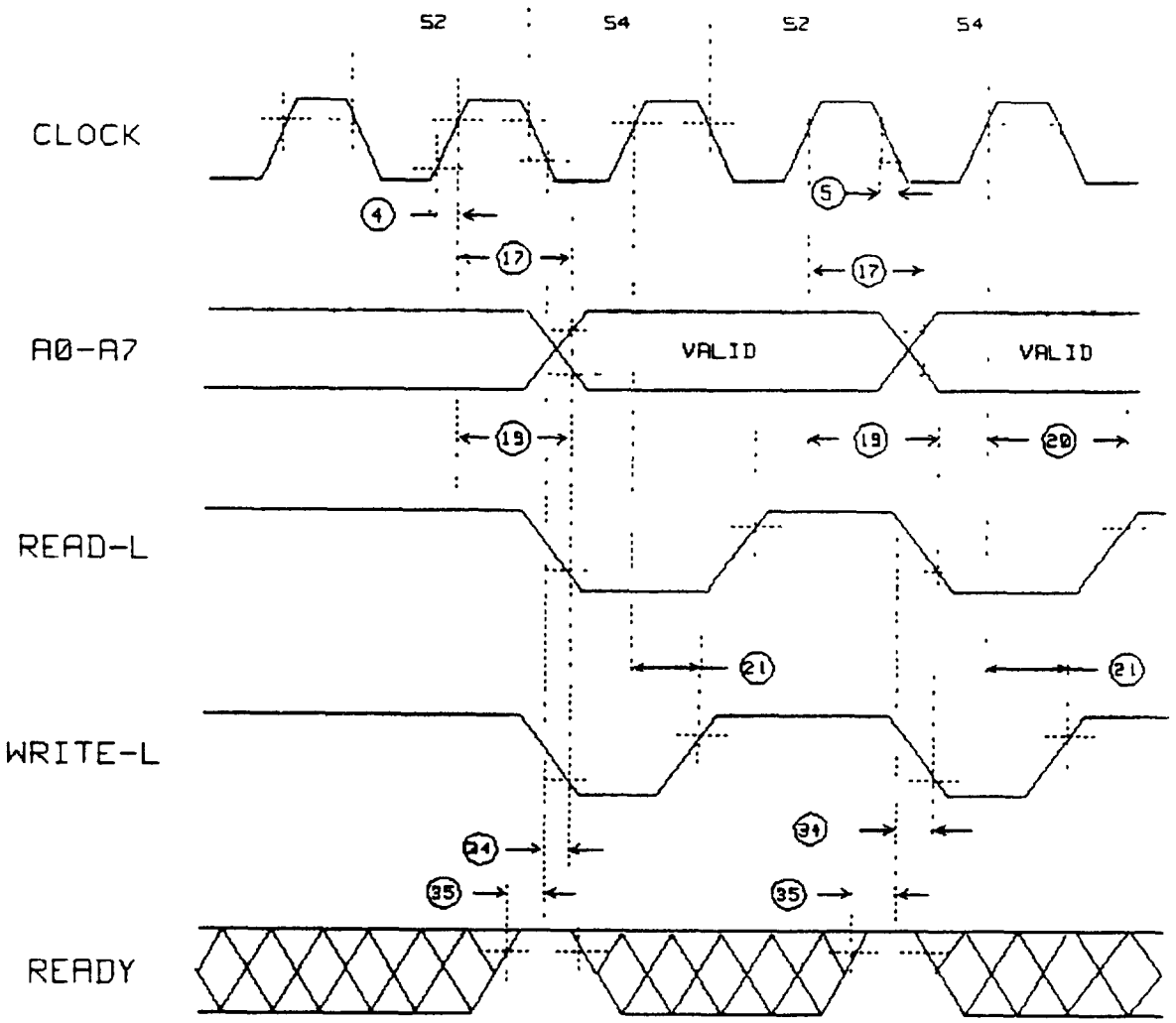


FIGURE 7f. Dynamic waveforms for table XIV.

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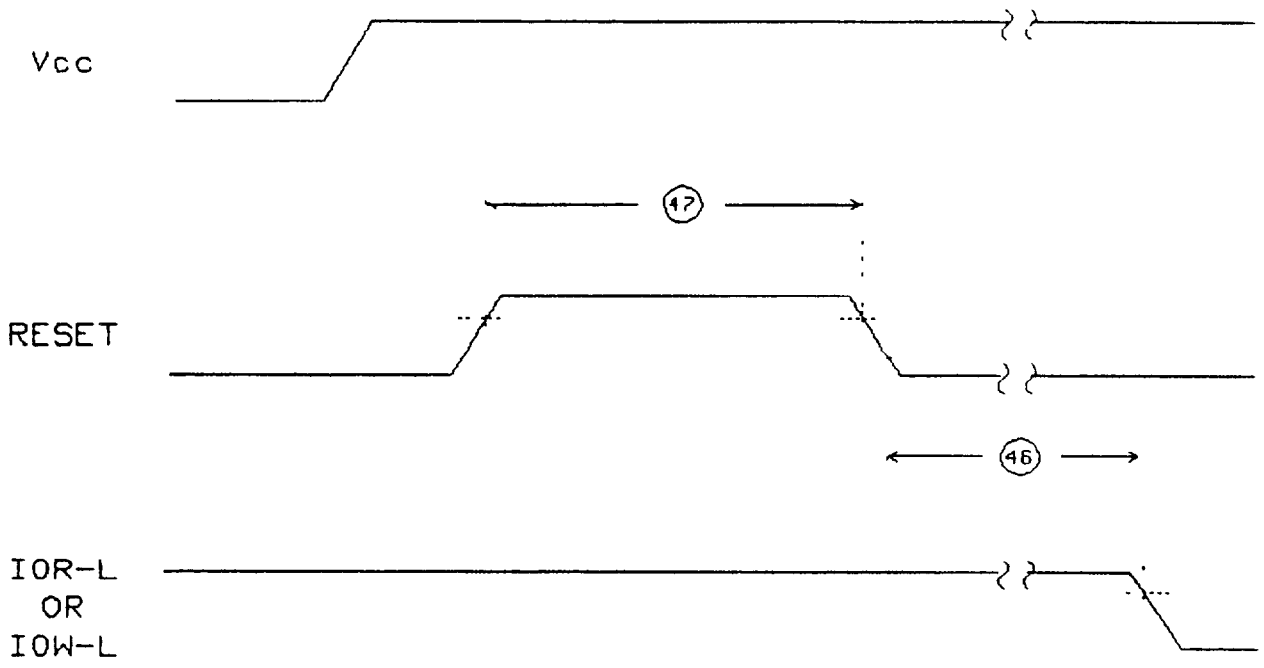


FIGURE 7g. Dynamic waveforms for table XIV.

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NOTES:

1. The new IOW-L or MEMW-L pulse width for normal write will be PRP-100 nS and for extended write will be 2PRP-100 nS. The net IOR-L or MEMR-L pulse width for normal read will be 2PRP-50 nS and for compressed read will be PRP-50 nS.

2. DREQ should be held active until DACK is returned.

3. DREQ and DACK signals may be active high or active low. Timing figures assume the active high mode.

4. Successive read and/or write operations by the external processor to program or examine the controller must be timed to allow at least 600 nS for the NBP as recovery time between active read or write pulses.

5. Signals READ-L and WRITE-L refer to IOR-L and MEMW-L, respectively, for peripheral-to-memory DMA operations and to MEMR-L and IOW-L, respectively, for memory-to-peripheral DMA operations.

6. If N wait states are added during the write-to-memory half of a memory-to-memory transfer, this parameter will increase by N(PRP).

7. The HREQ valid from CLOCK delay time is specified for two output high levels. Reference number 22a delay time is measured at 2.0 V. Reference number 22b delay time is measured at 3.3 V. The value for the 22b delay time assumes an external 3.32 KOhm pullup resistor connected from HREQ to Vcc.

8. Unless otherwise noted, waveform measurement points for both input and output signals are 2.0 V for high and 0.8 V for low.

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TABLE XV. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
N	PSFC	1	3.3	6.12	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	N
Capacitance bridge voltage	0.3 Vp	± 30 mVp	N
Capacitance measurement	NA	± 140 nF	N

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XV.

3. Abbreviations used are defined in 6.1.

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TABLE XVI. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
P1	A0	12	0	500	mV	1-4	8,9
P2	A1	13					
P3	A2	14					
P4	A3	15					
P5	A4	17					
P6	A5	18					
P7	A6	19					
P8	A7	20					
P9	HREQ	11					
P10	DACK0	6					
P11	DACK1	5					
P12	DACK2	27					
P13	DACK3	26					
P14	AEN	32					
P15	ADSTB	33					
P16	MEMR-L	38					
P17	MEMW-L	37					
P18	D0	21					
P19	D1	30					
P20	D2	9	0	500	mV	1-4	8,9

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TABLE XVI. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
P21	D3	8	0	500	mV	1-4	8,9
P22	D4	7					
P23	D5	4					
P24	D6	3					
P25	D7	2					
P26	IOR-L	40					
P27	IOW-L	39					
P28	EOP-L	16	0	500	mV	1-4	8,9

NOTES:

1. Output crosstalk, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	P1-P28
Power supply current limit	220 mA	NA	P1-P28
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 P1-P28 P1-P28
Output crosstalk measurement	NA	± 20 mV	P1-P28

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XVI, connect probe 'B' and visually inspect the output for the most negative spike as indicated on figure 8.

(e) The maximum probe capacitance is 10 pF.

3. The test circuit shall be as indicated on figure 9. All outputs shall be loaded as specified on figure 5.

4. Abbreviations used are defined in 6.1.

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TABLE XVII. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
Q1	A0	12	0	500	mV	1-4	8,9
Q2	A1	13					
Q3	A2	14					
Q4	A3	15					
Q5	A4	17					
Q6	A5	18					
Q7	A6	19					
Q8	A7	20					
Q9	HREQ	11					
Q10	DACK0	6					
Q11	DACK1	5					
Q12	DACK2	27					
Q13	DACK3	26					
Q14	AEN	32					
Q15	ADSTB	33					
Q16	MEMR-L	38					
Q17	MEMW-L	37					
Q18	D0	21					
Q19	D1	30					
Q20	D2	9	0	500	mV	1-4	8,9

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TABLE XVII. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
Q21	D3	8	0	500	mV	1-4	8,9
Q22	D4	7					
Q23	D5	4					
Q24	D6	3					
Q25	D7	2					
Q26	IOR-L	40					
Q27	IOW-L	39					
Q28	EOP-L	16	0	500	mV	1-4	8,9

NOTES:

1. Output crosstalk, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	Q1-Q28
Power supply current limit	220 mA	NA	Q1-Q28
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 Q1-Q28 Q1-Q28
Output crosstalk measurement	NA	± 20 mV	Q1-Q28

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XVII, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 8.

(e) The maximum probe capacitance is 10 pF.

3. The test circuit shall be as indicated on figure 9. All outputs shall be loaded as specified on figure 5.

4. Abbreviations used are defined in 6.1.

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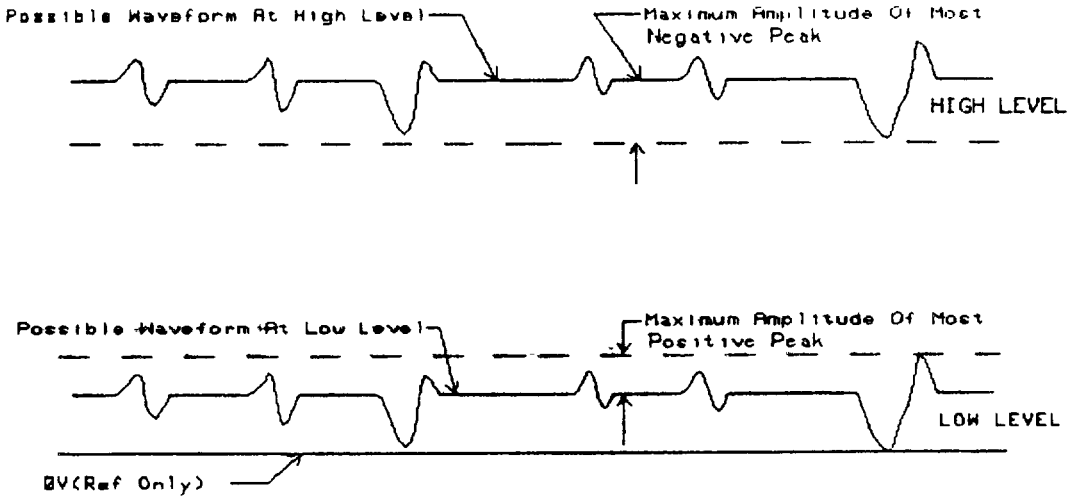


FIGURE 8. Output crosstalk, high and low level, test waveforms for tables XVI and XVII.

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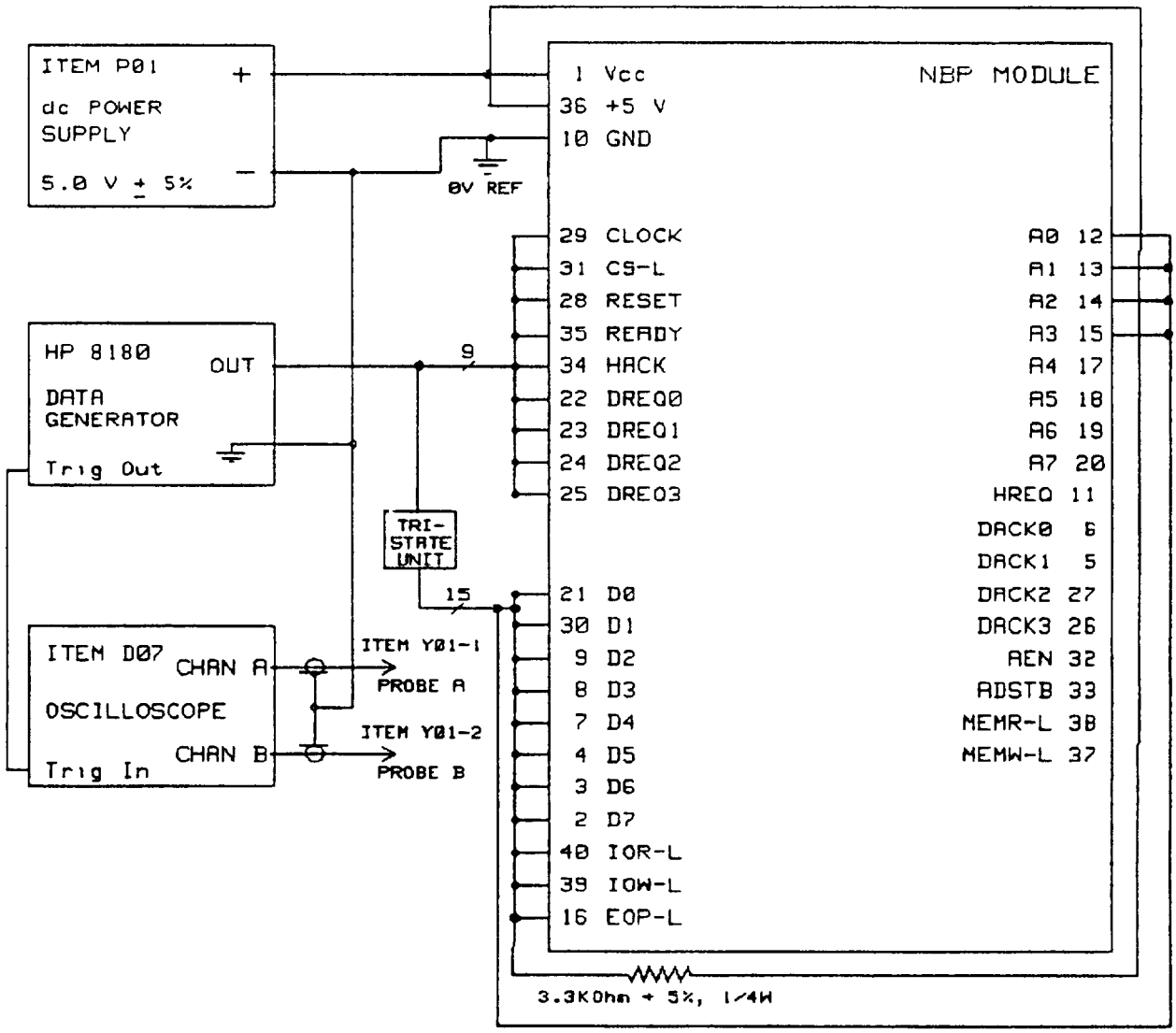


FIGURE 9. Output crosstalk, high and low level, test circuit for tables XVI and XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R	PSTA	1	0	500	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	R
Power supply current limit	220 mA	NA	R
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 R R
Power supply transient amplitude measurement	NA	± 20 mV	R

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XVIII.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XVIII.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 10 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 11.

4. Abbreviations used are defined in 6.1.

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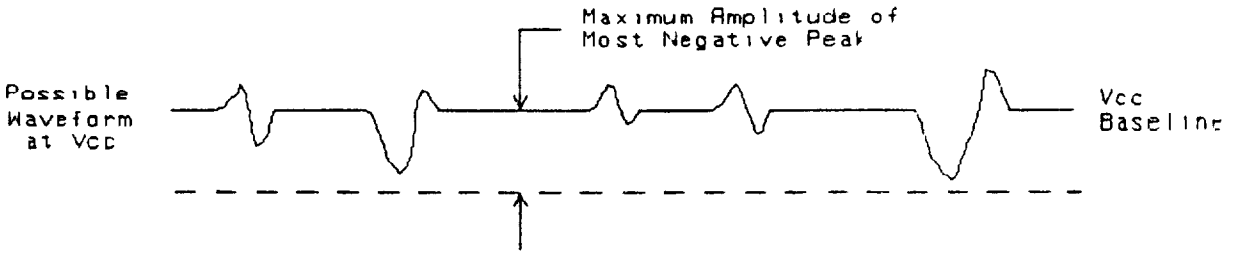


FIGURE 10. Power supply transient amplitude waveform for table XVIII.

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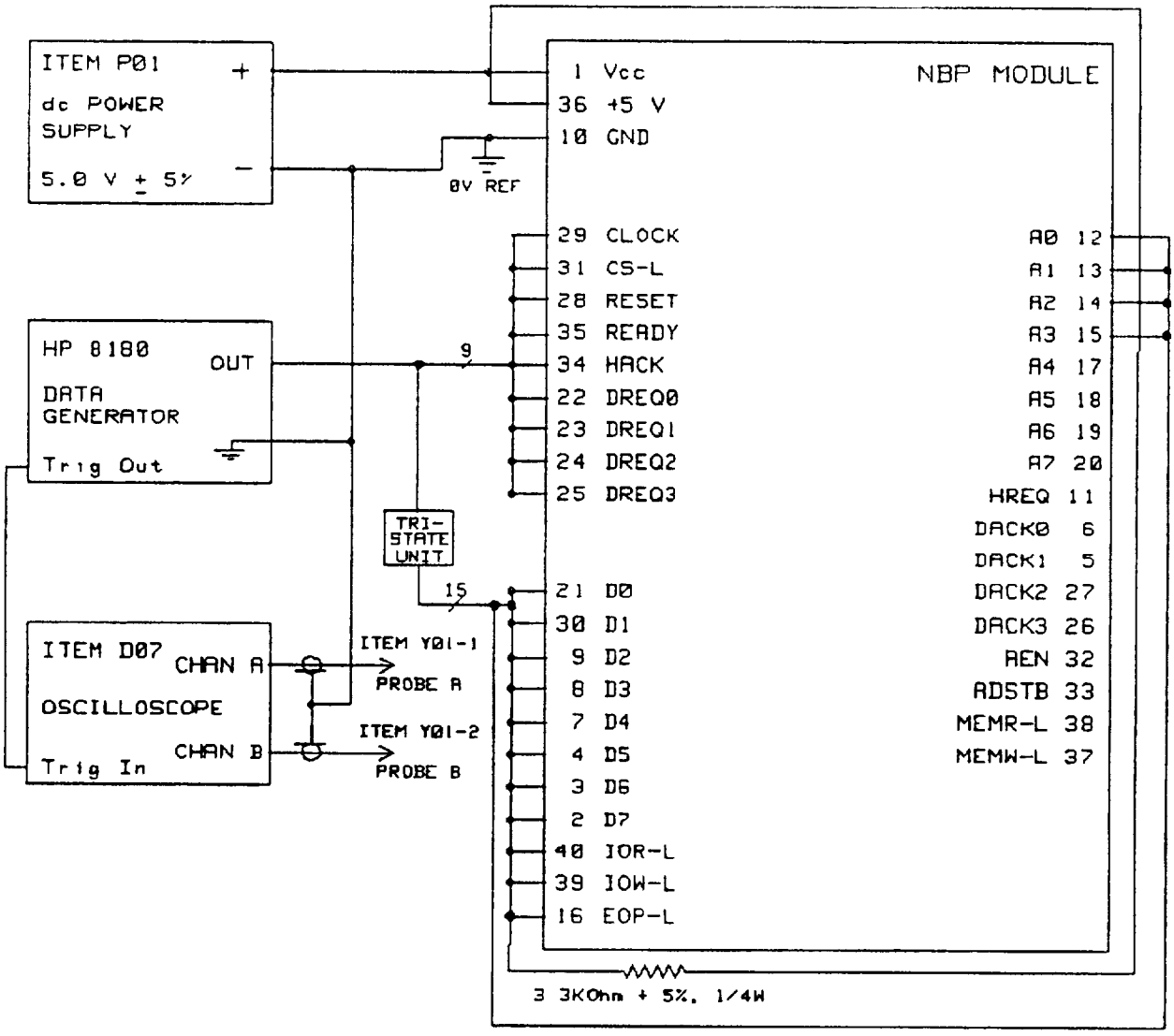


FIGURE 11. Power supply transient amplitude test circuit for table XVIII.

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TABLE XIX. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
S1	CLOCK	29	2	20	pF	1-3
S2	CS-L	31				
S3	RESET	28				
S4	READY	35				
S5	HACK	34				
S6	DREQ0	22				
S7	DREQ1	23				
S8	DREQ2	24				
S9	DREQ3	25	2	20	pF	1-3
S10	D0	21	2	25	pF	1-3
S11	D1	30				
S12	D2	9				
S13	D3	8				
S14	D4	7				
S15	D5	4				
S16	D6	3				
S17	D7	2				
S18	IOR-L	40				
S19	IOW-L	39				
S20	EOP-L	16	2	25	pF	1-3

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TABLE XIX. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
S21	A0	12	2	25	pF	1-3
S22	A1	13				
S23	A2	14				
S24	A3	15	2	25	pF	1-3
S25	A4	17	2	20	pF	1-3
S26	A5	18				
S27	A6	19				
S28	A7	20				
S29	HREQ	11				
S30	MEMR-L	38				
S31	MEMW-L	37	2	20	pF	1-3

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	S1-S31
Capacitance meter voltage	15 mV	± 1.5 mV	S1-S31
Capacitance measurement	NA NA NA	± 0.9 pF ± 1.15 pF ± 0.65 pF	S1-S9 S10-S24 S25-S31

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIX.

3. Abbreviations used are defined in 6.1.

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APPENDIX

10. SCOPE

10.1 Scope. This appendix contains information regarding the module function. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

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APPENDIX

30. Module function.

30.1 Major operating cycles. The NBP operates in two major cycles. These are called idle and active cycles. Each device cycle is made up of a number of states. The NBP can assume seven separate states, each composed of one full clock period. State 1 (S1) is the inactive state. It is entered when the NBP has no valid DMA requests pending. While in S1, the DMA controller is inactive but may be in the program condition, being programmed by the processor. State 0 (S0) is the first of a DMA service. The NBP has requested a hold but the processor has not yet returned an acknowledge. An ackowledge from the CPU will signal that transfers may begin. S1, S2, S3, and S4 are the working states of the DMA service. If more time is needed to complete a transfer than is available with normal timing, wait states (SW) can be inserted before S4 by the use of the ready line on the NBP.

30.1.1 Memory-to-memory transfers. Memory-to-memory transfers require a read-from and a write-to-memory to complete each transfer. The states, which resemble the normal working states, use two digit numbers for identification. Eight states are required for each complete transfer. The first four states (S11, S12, S13, S14) are used for the read-from-memory half and the last four states (S21, S22, S23, and S24) for the write-to-memory half of the transfer. The temporary data register is used for intermediate storage of the memory byte.

30.1.2 DMA service cycle. When no channel is requesting service, the NBP will enter the idle cycle and perform "S1" states. In this cycle the NBP will sample the DREQ lines every clock cycle to determine if any channel is requesting a DMA service. The device will also sample CS-L, looking for an attempt by the microprocessor to write or read the internal registers of the NBP. When CS-L is low and HACK is low, the NBP enters the program condition. The CPU can now establish, change or inspect the internal definition of the part by reading from or writing to the internal registers. Address lines A0-A3 are inputs to the device and select which registers will be read or written. The IOR-L and IOW-L lines are used to select and time reads or writes. Due to the number and size of the internal registers, an internal flip/flop is used to generate an additional bit of address. This bit is used to determine the upper or lower byte of the 16-bit address and word count registers. The flip/flop is reset by master clear or reset. A separate software command can also reset this flip/flop.

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30.1.3 Software commands. Special software commands can be executed by the NBP in the program condition. These commands are decoded as sets of addresses when both CS-L and IOW-L are active and do not make use of the data bus. Functions include clear first/last flip/flop and master clear.

30.2 TRANSFER TYPES. When the NBP is in the idle cycle and a channel requests a DMA service, the device will output a HREQ to the microprocessor and enter the active cycle. It is in this cycle that the DMA service will take place, in one of four modes.

30.2.1 Single transfer mode. In single transfer mode, the NBP will make a one-byte transfer during each HREQ/HACK handshake. When DREQ goes active, HREQ will go active. After the CPU responds by driving HACK active, a one-byte transfer will take place. Following the transfer, HREQ will go inactive, the word count will be decremented and the address will be either incremented or decremented. When the word count goes to zero, a terminal count (TC) will cause an autoinitialize if the channel has been programmed to do so.

30.2.1.1 Single transfer operation. To perform a single transfer, DREQ must be held active only until the corresponding DACK goes active. If DREQ is held continuously active, HREQ will go inactive following each transfer and then will go active again and a new one-byte transfer will be made following each rising edge of HACK. Details of timing between the NBP and other bus control protocols will depend upon the characteristics of the microprocessor involved.

30.2.2 Block transfer mode. In block transfer mode, the NBP will continue making transfers until a TC (caused by the word count going to zero) or an external end of process (EOP-L) is encountered. DREQ need be held active only until DACK becomes active. An autoinitialize will occur at the end of the service if the channel has been programmed for it.

30.2.3 Demand transfer mode. In demand transfer mode the device will continue making transfers until a TC or external EOP-L is encountered or until DREQ goes inactive. Thus, the device requesting service may discontinue transfers by bringing DREQ inactive. Service may be resumed by asserting an active DREQ once again. During the time between services when the microprocessor is allowed to operate, the intermediate values of

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address and word count may be read from the NBP current address and current word count registers. Autoinitialization will only occur following a TC or EOP-L at the end of service. Following autoinitialization, an active-going DREQ edge is required to initiate a new DMA service.

30.2.4 Cascade mode. This mode is used to cascade more than one NBP together for simple system expansion. The HREQ and HACK signals from the additional NBP are connected to the DREQ and DACK signals of a channel of the initial NBP. This allows the DMA requests of the additional device to propagate through the priority network circuitry of the preceding device. The priority chain is preserved and the new device must wait for its turn to acknowledge requests. Since the cascade channel in the initial device is used only for prioritizing the additional device, it does not output any address or control signals of its own. These would conflict with the outputs of the active channel on the added device. The NBP will respond to DREQ with DACK but all other outputs except HREQ will be disabled.

30.2.4.1 Transfer mode variations. Each of the three active transfer modes can perform three different types of transfers. These are read, write and verify. Write transfers move data from an I/O device to the memory by activating IOR-L and MEMW-L. Read transfers move data from memory to an I/O device by activating MEMW-L and IOW-L. Verify transfers are pseudo transfers; the NBP operates as in Read or Write transfers generating addresses, responding to EOP-L, etc., however, the memory and I/O control lines remain inactive.

30.2.5 Memory-to-memory. The NBP includes a block move capability that allows blocks of data to be moved from one memory address space to another. When bit C0 in the command register is set to a logical 1, channel 0 and 1 will operate as memory-to-memory transfer channels. Channel 0 forms the source address and channel 1 forms the destination address. The channel 1 word count is used. A memory-to-memory transfer is initiated by setting a software DMA request for channel 0. Block transfer mode should be used for memory-to-memory. When channel 0 is programmed for a fixed source address, a single source word may be written into a block of memory.

30.2.5.1 Memory-to-memory setup. When setting up the NBP for memmory-to-memory operation, it is suggested that both channels 0 and 1 be masked out. Further, the channel 0 word count should be initialized to the same value used in channel 1. No DACK outputs will be active during memory-to-memory transfers.

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30.2.5.2 Memory-to-memory termination. The NBP will respond to external EOP-L signals during memory-to-memory transfers. Data comparators in block search schemes may use this input to terminate the service when a match is found.

30.2.6 Autoinitialize. By programming a bit in the mode register a channel may be set up for an autoinitialize operation. During autoinitialization, the original values of the current address and current word count registers are automatically restored from the base address and base word count registers of that channel following EOP-L. The base registers are loaded simultaneously with the current registers by the microprocessor and remain unchanged throughout the DMA service. The mask bit is not set by EOP-L when the channel is in autoinitialize. Following autoinitialize the channel is ready to repeat its service without CPU intervention.

30.2.7 Priority fixed. The NBP has two types of priority encoding available as software selectable options. The first is fixed priority which fixes the channels in priority order based upon the descending value of their number. The channel with the lowest priority is 3 followed by 2,1 and the highest priority channel, 0.

30.2.7.1 Priority rotating. The second scheme is rotating priority. The last channel to get service becomes the lowest priority channel with the others rotating accordingly. With rotating priority in a single chip DMA system, any device requesting service is guaranteed to be recognized after no more than three higher priority services have occurred. This prevents any one channel from monopolizing the system.

30.2.7.2 Priority servicing. The priority encoder selects the highest priority channel requesting service on each active-going HACK edge. Once a channel is started, its operation will not be suspended if a request is received by a higher priority channel. The high priority channel will only gain control after the lower priority channel releases HREQ. When control is passed from one channel to another, the CPU will always gain bus control. This ensures generation of rising HACK edge to be used to initiate selection of the new highest-priority requesting channel.

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30.2.8 Compressed timing. In order to achieve even greater throughput where system characteristics permit, the NBP can compress the transfer time to two clock cycles. State S3 is used to extend the access time of the read pulse. By removing state S3, the read pulse width is made equal to the write pulse width and a transfer consists only of state S2 to change the address and state S4 to perform the read/write. S1 states will still occur when A8-A15 need updating (see address generation).

30.2.9 Address generation. In order to reduce pin count, the NBP multiplexes the eight higher order address bits on the data lines. State S1 is used to output the higher order address bits to an external latch from which they may be placed on the address bus. The falling edge of address strobe (ADSTB) is used to load these bits from the data lines to the latch. Address enable (AEN) is used to enable the bits onto the address bus through a 3-state enable. The lower order address bits are output by the NBP directly. Lines A0-A7 should be connected to the address bus.

30.2.9.1 Memory-to-memory addressing. During block and demand transfers mode services which include multiple transfers, the addresses generated will be sequential. For many transfers, the data held in the external address latch will remain the same. This data need only change when a carry or borrow from A7 to A8 takes place in the normal sequence of addresses. To save time and speed transfers, the NBP executes S1 states only when updating of A8-A15 in the latch is necessary. This means for long services, 4 states may occur only once every 256 transfers, a savings of 255 clock cycles for each 256 transfers.

30.3 REGISTER DESCRIPTION

30.3.1 Current address register. Each channel has a 16-bit current address register. This register holds the value of the address used during DMA transfers. The address is automatically incremented or decremented after each transfer and the intermediate values of the address are stored in the current address register during the transfer. This register is written or read by the microprocessor in successive 8-bit bytes. It may also be reinitialized by an autoinitialize back to its original value. Autoinitialization takes place only after an EOP-L.

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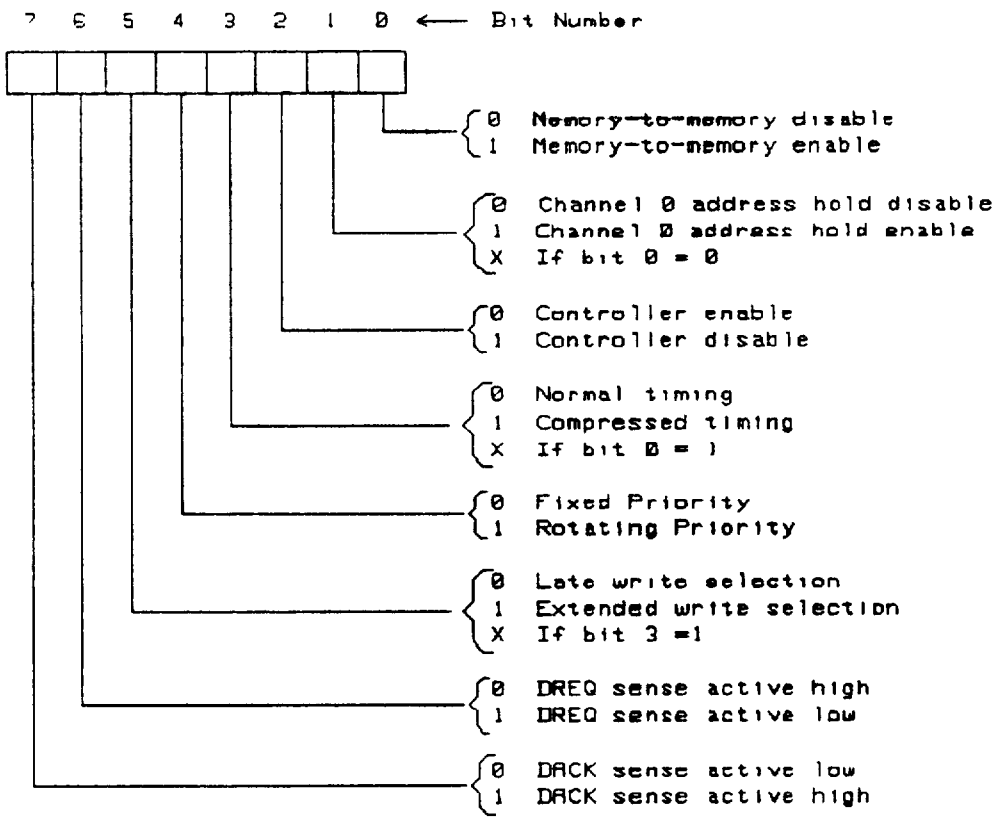
30.3.2 Current word count register. Each channel has a 16-bit current word count register. This register should be programmed with, and will return on a CPU read, a value one less than the number of words to be transferred. The word count is decremented after each transfer. The intermediate value of the word count is stored in the register during the transfer. When the value in the register goes to zero, a TC will be generated. This register is loaded or read in successive 8-bit bytes by the microprocessor in the program condition. Following the end of a DMA service, it may also be reinitialized by an autoinitialize back to its original value. Autoinitialize can occur only when an EOP-L occurs. Note that the contents of the word count register will be FFFF (hex) following an internally generated EOP-L.

30.3.3 Base address and base word count registers. Each channel has a pair of base address and base word count registers. These 16-bit registers store the original values of their associated current registers. During autoinitialize, these values are used to restore the current registers to their original values. The base registers are written simultaneously with their corresponding current register in 8-bit bytes during DMA programming by the microprocessor. Accordingly, writing to these registers when intermediate values are in the current registers will overwrite the intermediate values. The base registers cannot be read by the microprocessor.

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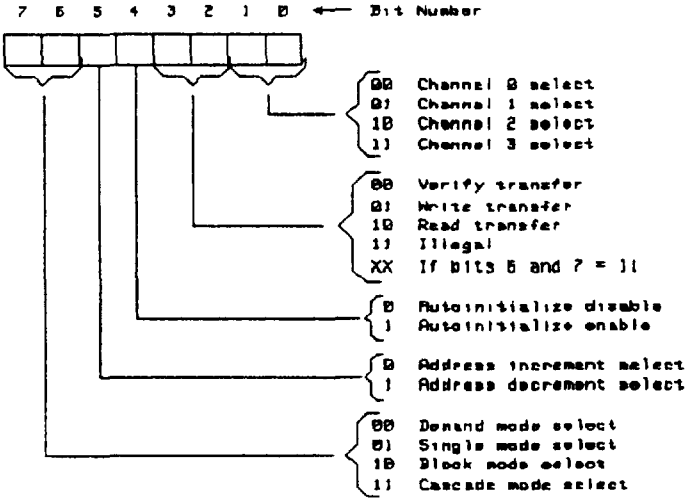
APPENDIX

30.3.4 Command register. This 8-bit register controls the operation of the NBP. It is programmed by the microprocessor in the program condition and is cleared by reset. The following table lists the function of the command bits.

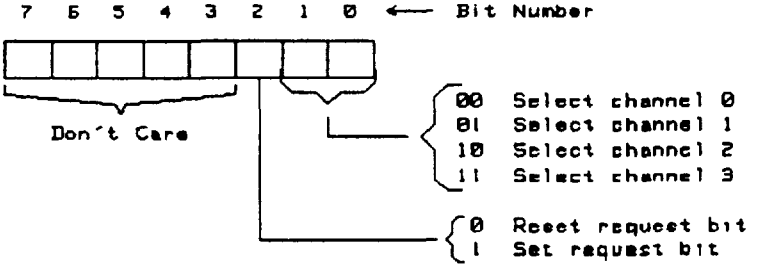


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30.3.5 Mode register. Each channel has a 6-bit mode register associated with it. When the register is being written to by the microprocessor in the program condition, bits 0 and 1 determine which channel mode register is to be written.



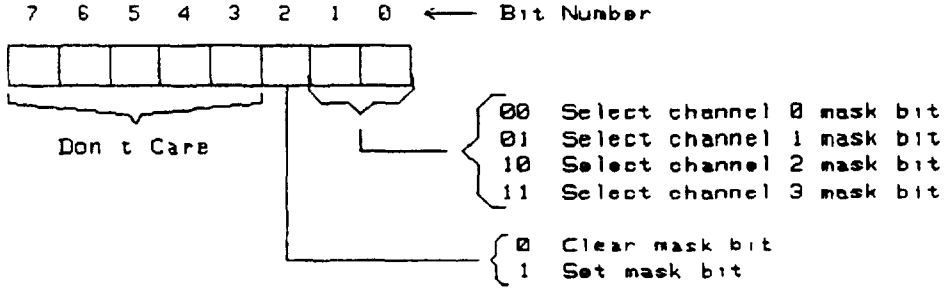
30.3.6 Request register. The NBP can respond to requests for DMA services which are initiated by software as well as by a DREQ. Each channel has a request bit associated with it in the 4-bit request register. These are nonmaskable and subject to prioritization by the priority encoder network. Each register bit is set or reset separately under software control or is cleared upon generation of a TC or external EOP. The entire register is cleared by a reset. To set or reset a bit, the software loads the proper form of the data word.



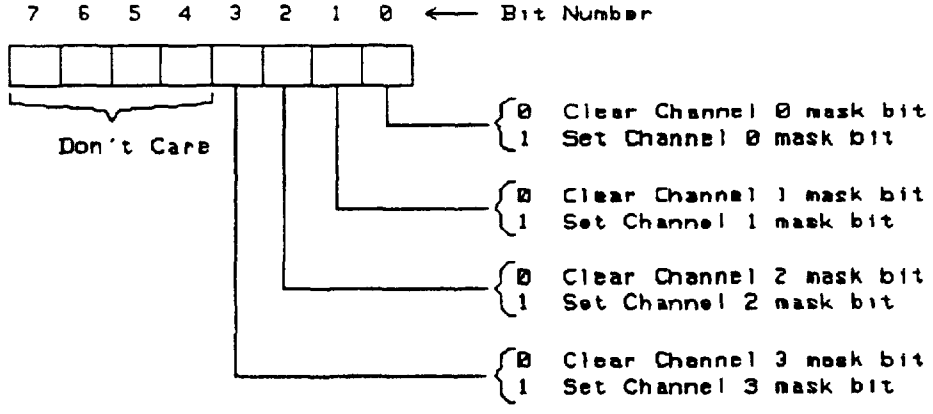
Software requests will be serviced only if the channel is in block mode. When initiating a memory-to-memory transfer, the software request for channel 0 should be set.

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30.3.7 Mask register. Each channel has associated with it a mask bit which can be set to disable the incoming DREQ. Each mask bit is set when its associated channel produces an EOP if the channel is not programmed for autoinitialize. Each bit of the 4-bit mask register may also be set or cleared separately under software control. The entire register is also set by a reset. This disables all DMA requests until a clear mask register instruction allows them to occur. The instruction to separately set or clear the mask bits is similar in form to that used with the request register.

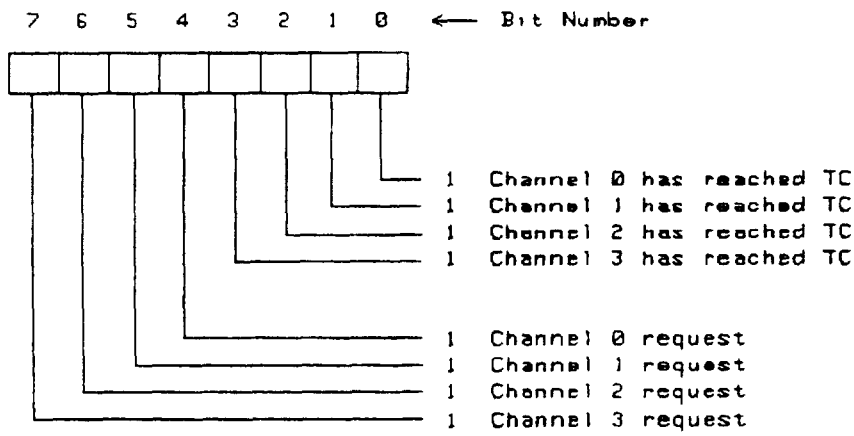


All four bits of the mask register may also be written in a single command.



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30.3.8 Status register. The status registers may be read out of the NBP by the microprocessor. It indicates which channels have reached a terminal count and which channels have pending DMA requests. Bits 0-3 are set each time a TC is reached by that channel, including after each autoinitialization. These bits are cleared by reset and each status read. Bits 4-7 are set whenever their corresponding channel is requesting service.



30.3.9 Temporary register. The temporary register is used to hold data during memory transfers. Following the completion of the transfers, the last word moved can be read by the microprocessor in the program condition. The temporary register always contains the last byte transferred in the previous memory-to-memory operation, unless cleared by a reset.

30.3.10 Software commands. There are two special software commands which can be executed in the program condition. They do not depend on any specific bit pattern on the data bus. The two software commands are:

30.3.10.1 Clear First/Last Flip/Flop. This command may be issued prior to writing or reading NBP address or word count information. This initializes the flip/flop to a known state so that subsequent accesses to register contents by the microprocessor will address lower and upper bytes in the correct sequence.

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30.3.10.2 Master clear. This software instruction has the same effect as the hardware reset. The command, status, request, temporary and internal first/last flip/flop registers are cleared and the mask register is set. The NBP will enter the idle cycle. The following tables show the register and function addressing, and the word count and address registration command codes.

TABLE I. Register and function addressing.

Interface Signal						OPERATION
A3	A2	A1	A0	IOR-L	IOW-L	
1	0	0	0	0	1	Read status register
1	0	0	0	1	0	Write command register
1	0	0	1	0	1	Illegal
1	0	0	1	1	0	Write request register
1	0	1	0	0	1	Illegal
1	0	1	0	1	0	Write single mask register bit
1	0	1	1	0	1	Illegal
1	0	1	1	1	0	Write mode register
1	1	0	0	0	1	Illegal
1	1	0	0	1	0	Clear byte pointer flip/flop
1	1	0	1	0	1	Read temporary register
1	1	0	1	1	0	Master clear
1	1	1	0	0	1	Illegal
1	1	1	0	1	0	Illegal
1	1	1	1	0	1	Illegal
1	1	1	1	1	0	Write all mask register bits

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TABLE II. Word count and address register command codes.

Chan.	Register	Op.	Signals							Internal flip/flop	Data bus D0-D7
			CS-L	IOR-L	IOW-L	A3	A2	A1	A0		
0	Base & current address	Write	0	1	0	0	0	0	0	0	A0-A7
			0	1	0	0	0	0	0	1	A8-A15
	Current address	Read	0	0	1	0	0	0	0	0	A0-A7
			0	0	1	0	0	0	0	1	A8-A15
	Base & current word count	Write	0	1	0	0	0	0	1	0	W0-W7
			0	1	0	0	0	0	1	1	W8-W15
	Current word count	Read	0	0	1	0	0	0	1	0	W0-W7
			0	0	1	0	0	0	1	1	W8-W15
1	Base & current address	Write	0	1	0	0	0	1	0	0	A0-A7
			0	1	0	0	0	1	0	1	A8-A15
	Current address	Read	0	0	1	0	0	1	0	0	A0-A7
			0	0	1	0	0	1	0	1	A8-A15
	Base & current word count	Write	0	1	0	0	0	1	0	0	W0-W7
			0	1	0	0	0	1	0	1	W8-W15
	Current word count	Read	0	0	1	0	0	1	1	0	W0-W7
			0	0	1	0	0	1	1	1	W8-W15

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TABLE II. Word count and address register command codes - Continued.

Chan.	Register	Op.	Signals							Internal flip/flop	Data bus D0-D7
			CS-L	IOR-L	IOW-L	A3	A2	A1	A0		
2	Base & current address	Write	0	1	0	0	1	0	0	0	A0-A7
			0	1	0	0	1	0	0	1	A8-A15
	Current address	Read	0	0	1	0	1	0	0	0	A0-A7
			0	0	1	0	1	0	0	1	A8-A15
	Base & current word count	Write	0	1	0	0	1	0	1	0	W0-W7
			0	1	0	0	1	0	1	1	W8-W15
	Current word count	Read	0	0	1	0	1	0	1	0	W0-W7
			0	0	1	0	1	0	1	1	W8-W15
3	Base & current address	Write	0	1	0	0	1	1	0	0	A0-A7
			0	1	0	0	1	1	0	1	A8-A15
	Current address	Read	0	0	1	0	1	1	0	0	A0-A7
			0	0	1	0	1	1	0	1	A8-A15
	Base & current word count	Write	0	1	0	0	1	1	1	0	W0-W7
			0	1	0	0	1	1	1	1	W8-W15
	Current word count	Read	0	0	1	0	1	1	1	0	W0-W7
			0	0	1	0	1	1	1	1	W8-W15

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